A right triangle has vertices (0, 0), (8, 0), (0, 6). Circumradius?

A. 5

B. 10

C. 4

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q4/5

2401

QUESTION ID

S60-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 5

GRADING NOTE Half the hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how to check a computed circumcenter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q5/5

2402

QUESTION ID

S60-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Verify it is the same distance from all three vertices.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 60 • FORWARD 1 • Q5/5

SET 60 • HSG-C.A.3

FORWARD 2

Find the circumradius with trigonometry (extended Law of Sines) • High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10

ANSWER KEY Reduced copy of the question page, with its answer

In a triangle, a = 10 and A = 30°. Find the circumradius R using a/sin A = 2R.

R = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q1/5

2404

QUESTION ID
S60-F2-Q1

TYPE
Short answer

CORRECT ANSWER
10

GRADING NOTE Any equivalent form. 10 / 0.5 = 20 = 2R.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Equilateral triangle with side 6. Find the circumradius exactly.

R = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q2/5

2405

QUESTION ID
S60-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$2\sqrt{3}$

GRADING NOTE Form required: exact. $6/\sin 60° = 4\sqrt{3} = 2R$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q2/5

True or false?

In any triangle, a/sin A equals the radius of the circumcircle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S60-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q3/5

2406

QUESTION ID

S60-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It equals the diameter, 2R.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S60-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q3/5

A triangle with a 90° angle opposite side 14 has circumradius

A. 3.5

B. 14

C. 28

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q4/5

2407

QUESTION ID

S60-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 7

GRADING NOTE 14/sin 90° = 14 = 2R.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q4**

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why a/sin A = 2R using an inscribed angle on a diameter.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q5/5

2408

GRADING NOTE Any equivalent explanation.

QUESTION ID

S60-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Draw the diameter from B; the inscribed angle at its far end equals A (same arc) and is in a right triangle with hypotenuse 2R, so sin A = a/(2R).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.D.10 • Set 60 • FORWARD 2 • Q5/5

SET 61 • HSG-C.A.3 • CIRCLES

Prove that opposite angles of an inscribed quadrilateral are supplementary

MAIN	Geometry main question	**HSG-C.A.3**
BACKWARD 1	The inscribed angle theorem	**HSG-C.A.2**
BACKWARD 2	Supplementary angles	**7.G.B.5**
BACKWARD 3	Arcs of a circle add to 360°	**4.MD.C.7**
FORWARD 1	Use the converse: decide whether a quadrilateral can be inscribed	**HSG-C.A.3**
FORWARD 2	Exterior angle of a cyclic quadrilateral	**HSG-C.A.3**

SET 61 • HSG-C.A.3

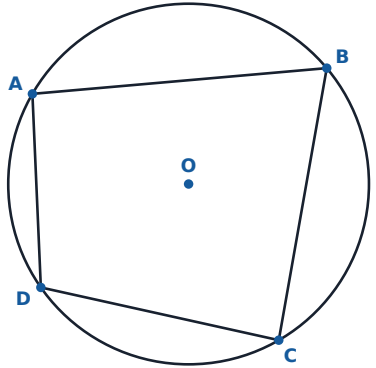
MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Quadrilateral ABCD is inscribed in circle O. Prove that m∠A + m∠C = 180°.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q1/5

2411

QUESTION ID

S61-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠A + m∠C = 180°

GRADING NOTE REQUIRED WORK: Use the inscribed angle theorem: ∠A and ∠C intercept the two arcs BCD and BAD, which together make the whole circle.

Any valid proof (paragraph, two-column, or flow). Essential: m∠A = ½ arc BCD and m∠C = ½ arc BAD (inscribed angle theorem); the two arcs add to 360°; so the angles add to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Quadrilateral PQRS is inscribed in a circle. Prove that ∠Q and ∠S are supplementary.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q2/5

2412

QUESTION ID

S61-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$m\angle Q + m\angle S = 180°$

GRADING NOTE REQUIRED WORK: Use the inscribed angle theorem: ∠Q and ∠S intercept arcs PSR and PQR, which together make the whole circle.

Any valid proof (paragraph, two-column, or flow). Essential: each angle is half its intercepted arc, and the two intercepted arcs make up the whole circle (360°).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Quadrilateral ABCD is inscribed in a circle with m∠A = 72° and m∠B = 105°. Find m∠C and m∠D, justifying each with the inscribed-quadrilateral theorem.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q3/5

2413

QUESTION ID

S61-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

m∠C = 108°,
m∠D = 75°

GRADING NOTE REQUIRED WORK: Use that opposite angles of an inscribed quadrilateral are supplementary.

Any valid proof (paragraph, two-column, or flow). Essential: ∠A and ∠C are opposite, so m∠C = 180° − 72° = 108°; ∠B and ∠D are opposite, so m∠D = 180° − 105° = 75°; the theorem named for each.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q3/5

Parallelogram ABCD is inscribed in a circle. Prove that ABCD is a rectangle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q4/5

2414

QUESTION ID

S61-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

ABCD is a rectangle.

GRADING NOTE REQUIRED WORK: Opposite angles of a parallelogram are congruent; opposite angles of an inscribed quadrilateral are supplementary.

Any valid proof (paragraph, two-column, or flow). Essential: ∠A ≅ ∠C (parallelogram) and m∠A + m∠C = 180° (inscribed), so each is 90°; a parallelogram with a right angle is a rectangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q4/5

Trapezoid ABCD with AB ∥ DC is inscribed in a circle. Prove that ∠A ≅ ∠B (so the trapezoid is isosceles).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q5/5

2415

QUESTION ID

S61-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

∠A ≅ ∠B

GRADING NOTE REQUIRED WORK: Compare same-side interior angles (from the parallel sides) with opposite angles (from the circle).

Any valid proof (paragraph, two-column, or flow). Essential: m∠A + m∠D = 180° (same-side interior angles, AB ∥ DC); m∠B + m∠D = 180° (opposite angles of an inscribed quadrilateral); so m∠A = m∠B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • MAIN • Q5/5

SET 61 • HSG-C.A.3

BACKWARD 1

The inscribed angle theorem • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

An inscribed angle intercepts a 220° arc. Its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q1/5

2417

QUESTION ID

S61-B1-Q1

TYPE

Short answer

CORRECT ANSWER

110°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q1/5

True or false?

An inscribed angle can intercept an arc greater than 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q2/5

2418

QUESTION ID

S61-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Then the angle is obtuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q2/5

An inscribed angle of 70° intercepts

A. 70°

B. 290°

C. 35°

D. 140°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q3/5

2419

QUESTION ID

S61-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 140°

GRADING NOTE Double.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two arcs together make a whole circle. Their sum?

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q4/5

2420

QUESTION ID

S61-B1-Q4

TYPE

Short answer

CORRECT ANSWER

360°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q4/5

True or false?

An inscribed angle equals half the central angle only when it is acute.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q5/5

2421

QUESTION ID

S61-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Always.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • HSG-C.A.3

BACKWARD 2

Supplementary angles • Grade 7 • Geometry • 7.G.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Supplement of 72°?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q1/5

2423

QUESTION ID

S61-B2-Q1

TYPE

Short answer

CORRECT ANSWER

108°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q1/5

True or false?

Two angles whose measures add to 180° are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q2/5

2424

QUESTION ID

S61-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Angles 3x and 2x are supplementary. x =

A. 30°

B. 45°

C. 36°

D. 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q3/5

2425

QUESTION ID

S61-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 36°

GRADING NOTE 5x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If ∠A = ∠C and they are supplementary, find each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q4/5

2426

QUESTION ID

S61-B2-Q4

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Supplementary angles are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q5/5

2427

QUESTION ID

S61-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only if both are 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 61 • BACKWARD 2 • Q5/5

SET 61 • HSG-C.A.3

BACKWARD 3

Arcs of a circle add to 360° • Grade 4 • Measurement & Data • 4.MD.C.7

ANSWER KEY Reduced copy of the question page, with its answer

A circle is split into arcs of 100°, 80°, and x°. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q1/5

2429

QUESTION ID

S61-B3-Q1

TYPE

Short answer

CORRECT ANSWER

180

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q1/5

True or false?

The arcs between four points on a circle add to 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q2/5

2430

QUESTION ID

S61-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Whole circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q2/5

Two arcs make a whole circle; one is 150°. The other is

A. 210°

B. 150°

C. 30°

D. 360°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q3/5

2431

QUESTION ID

S61-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 210°

GRADING NOTE 360 − 150.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Half of 360° is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q4/5

2432

QUESTION ID

S61-B3-Q4

TYPE

Short answer

CORRECT ANSWER

180°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q4/5

True or false?

The arcs of a circle add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q5/5

2433

QUESTION ID

S61-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 360°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 61 • BACKWARD 3 • Q5/5

SET 61 • HSG-C.A.3

FORWARD 1

Use the converse: decide whether a quadrilateral can be inscribed • High School — Geometry • Geometry: Circles • HSG-C.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Quadrilateral ABCD has angles A = 85°, B = 70°, C = 95°, D = 110°. Can it be inscribed in a circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q1/5

2435

QUESTION ID

S61-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: A + C = 180° and B + D = 180°.

GRADING NOTE Form required: the conclusion with the check. (The converse is true: such a quadrilateral is cyclic.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q1/5

An isosceles trapezoid has base angles of 64°. Can it be inscribed in a circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q2/5

2436

QUESTION ID

S61-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes: its other angles are 116°, and 64° + 116° = 180° for each pair of opposite angles.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q2/5

True or false?

Every kite can be inscribed in a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q3/5

2437

QUESTION ID

S61-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only kites with two right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q3/5

Inscribed quadrilateral with angles (2x)°, (3x + 5)°, (2x + 20)°, and y°. If 2x and 2x + 20 are opposite, x =

A. 40

B. 50

C. 45

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q4/5

2438

QUESTION ID

S61-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 40

GRADING NOTE 4x + 20 = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Inscribed quadrilateral: ∠A = (5y − 10)°,
∠C = (3y + 30)°, A and C opposite. Find y and ∠A.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q5/5

2439

QUESTION ID
S61-F1-Q5

TYPE
Short answer

CORRECT ANSWER
y = 20; ∠A = 90°

GRADING NOTE Form required: both. 8y + 20 = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 1 • Q5/5

SET 61 • HSG-C.A.3

FORWARD 2

Exterior angle of a cyclic quadrilateral • High School — Geometry • Geometry: Circles • HSG-C.A.3

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is inscribed in a circle and side DC is extended past C to E. Prove that m∠BCE = m∠A.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q1/5

2441

QUESTION ID

S61-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠BCE = m∠A

GRADING NOTE REQUIRED WORK: Use the linear pair at C and the supplementary opposite angles.

Any valid proof (paragraph, two-column, or flow). Essential: m∠BCE = 180° − m∠BCD (linear pair); m∠A = 180° − m∠BCD (opposite angles); so equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In inscribed ABCD, ∠A = 77°. Side DC is extended past C to E. Find ∠BCE.

∠BCE = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q2/5

2442

QUESTION ID

S61-F2-Q2

TYPE

Short answer

CORRECT ANSWER

77°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q2/5

True or false?

An exterior angle of a cyclic quadrilateral equals the interior angle next to it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q3/5

2443

QUESTION ID

S61-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is supplementary to the adjacent interior angle and equals the opposite one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Inscribed ABCD with ∠B = 100°. The exterior angle at D (extending CD past D) is

A. 50°

B. 180°

C. 100°

D. 80°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q4/5

2444

QUESTION ID

S61-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 100°

GRADING NOTE Equals the opposite interior angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q4/5

ABCD is inscribed in a circle, and side DC is extended past C. Why is the exterior angle at C equal to ∠A, not ∠B?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q5/5

2445

QUESTION ID

S61-F2-Q5

TYPE

Short answer

CORRECT ANSWER

It is supplementary to ∠C, and the angle supplementary to ∠C is its opposite angle, ∠A.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 61 • FORWARD 2 • Q5/5

SET 62 • HSG-C.B.5 • CIRCLES

Derive arc length from similarity and define radian measure

MAIN	Geometry main question	**HSG-C.B.5**
BACKWARD 1	Similar figures: lengths scale by the scale factor	**HSG-SRT.A.2**
BACKWARD 2	Circumference of a circle	**7.G.B.4**
BACKWARD 3	Find a fraction of a whole	**7.RP.A.3**
FORWARD 1	Convert between degrees and radians	**HSG-C.B.5**
FORWARD 2	Radian measure for angles of rotation and the unit circle	**HSF-TF.A.1**

SET 62 • HSG-C.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A central angle θ cuts off arc length s_1 on a circle of radius r_1 and arc length s_2 on a circle of radius r_2 (same center). Explain, using similarity, why $s_1/r_1 = s_2/r_2$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q1/5

2448

QUESTION ID

S62-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$s_2 = (r_2/r_1)s_1$, so $s_1/r_1 = s_2/r_2$.

GRADING NOTE REQUIRED WORK: Use the dilation about the center with factor r_2/r_1: it maps the first circle to the second and the first arc to the second arc.

Any valid proof (paragraph, two-column, or flow). Essential: the dilation about the center with factor $k = r_2/r_1$ maps the first circle onto the second and the arc of angle θ onto the arc of angle θ; it multiplies all lengths, including arc length, by k; so $s_2 = k\, s_1$ and s/r is the same for both. This constant is θ in radians.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q1/5

A central angle cuts off an arc of length 12 cm on a circle of radius 4 cm. What is the angle in radians?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q2/5

2449

QUESTION ID
S62-M-Q2

TYPE
Short answer

CORRECT ANSWER
3 radians

GRADING NOTE Any equivalent form. θ = s/r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the length of the arc cut off by a central angle of 2.5 radians in a circle of radius 6 m.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q3/5

2450

QUESTION ID

S62-M-Q3

TYPE

Short answer

CORRECT ANSWER

15 m

GRADING NOTE Any equivalent form. s = rθ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the exact length of the arc cut off by a 120° central angle in a circle of radius 9.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q4/5

2451

QUESTION ID

S62-M-Q4

TYPE

Short answer

CORRECT ANSWER

6π

GRADING NOTE Form required: exact value. 120° = 2π/3 radians; s = 9 · 2π/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use s = rθ to explain why a full turn is 2π radians and a half turn is π radians, and convert 270° to radians.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q5/5

2452

QUESTION ID

S62-M-Q5

TYPE

Short answer

CORRECT ANSWER

A full turn cuts off the whole circumference 2πr, so θ = 2πr/r = 2π; half a turn cuts off πr, so θ = π; 270° is ¾ of a turn, 3π/2 radians.

GRADING NOTE Form required: both explanations and the conversion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • MAIN • Q5/5

SET 62 • HSG-C.B.5

BACKWARD 1

Similar figures: lengths scale by the scale factor • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

True or false?

Under a dilation with factor k, every length (straight or curved) is multiplied by k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q1/5

2454

QUESTION ID
S62-B1-Q1

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two similar circles have radii 3 and 12. Scale factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q2/5

2455

QUESTION ID

S62-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-B1-Q2**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q2/5

An arc of length 5 on the small circle (radius 3) matches which arc length on the large one (radius 12)?

A. 20

B. 15

C. 8

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q3/5

2456

QUESTION ID

S62-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 20

GRADING NOTE × 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Are all circles similar?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q4/5

2457

QUESTION ID

S62-B1-Q4

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q4/5

True or false?

Arc lengths do not change under dilation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q5/5

2458

QUESTION ID

S62-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 62 • BACKWARD 1 • Q5/5

SET 62 • HSG-C.B.5

BACKWARD 2

Circumference of a circle • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the circumference of a circle with radius 5 (exact).

Circumference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q1/5

2460

QUESTION ID

S62-B2-Q1

TYPE

Short answer

CORRECT ANSWER

10π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q1/5

True or false?

$C = 2\pi r$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q2/5

2461

QUESTION ID

S62-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q2/5

A circle has diameter 8. Its circumference is

A. 64π

B. 4π

C. 16π

D. 8π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q3/5

2462

QUESTION ID

S62-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 8π

GRADING NOTE πd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has circumference 18π. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q4/5

2463

QUESTION ID

S62-B2-Q4

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q4/5

True or false?

$C = \pi r^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q5/5

2464

QUESTION ID

S62-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That is area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 62 • BACKWARD 2 • Q5/5

SET 62 • HSG-C.B.5

BACKWARD 3

Find a fraction of a whole • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

What fraction of a circle is a 90° arc?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q1/5

2466

QUESTION ID

S62-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1/4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find ⅓ of 18π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q2/5

2467

QUESTION ID

S62-B3-Q2

TYPE

Short answer

CORRECT ANSWER

6π

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

An arc of 45° is what fraction of the circumference?

A. 45/100

B. ⅛

C. 1/45

D. ¼

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q3/5

2468

QUESTION ID

S62-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. ⅛

GRADING NOTE 45/360.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q3/5

True or false?

A 120° arc is ⅓ of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q4/5

2469

QUESTION ID

S62-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 120/360.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q4/5

True or false?

A 60° arc is 1/60 of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q5/5

2470

QUESTION ID

S62-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 3 • Q5/5

SET 62 • HSG-C.B.5

FORWARD 1

Convert between degrees and radians • High School — Geometry • Geometry: Circles • HSG-C.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Convert 150° to radians (exact).

Radians: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q1/5

2472

QUESTION ID

S62-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5π/6

GRADING NOTE Form required: exact value. Multiply by π/180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Convert 3π/4 radians to degrees.

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q2/5

2473

QUESTION ID

S62-F1-Q2

TYPE

Short answer

CORRECT ANSWER

135°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

1 radian is about 3.14°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q3/5

2474

QUESTION ID

S62-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1 radian ≈ 57.3°; π radians = 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q3/5

Which equals π/3 radians?

A. 60°

B. 120°

C. 30°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q4/5

2475

QUESTION ID

S62-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 60°

GRADING NOTE 180/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

An arc in a circle of radius 10 has length 5π. Find the central angle in degrees.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q5/5

2476

QUESTION ID

S62-F1-Q5

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form. θ = π/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 62 • FORWARD 1 • Q5/5

SET 62 • HSG-C.B.5

FORWARD 2

Radian measure for angles of rotation and the unit circle • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1

ANSWER KEY Reduced copy of the question page, with its answer

On the unit circle, an angle of 2 radians cuts off an arc of what length?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q1/5

2478

QUESTION ID
S62-F2-Q1

TYPE
Short answer

CORRECT ANSWER
2

GRADING NOTE Any equivalent form. r = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A wheel of radius 0.3 m rotates about its fixed center (axle) through 10 radians. How far does a point on its rim travel along its circular path?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q2/5

2479

QUESTION ID
S62-F2-Q2

TYPE
Short answer

CORRECT ANSWER
3 m

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

An angle of 6 radians is more than one full turn.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q3/5

2480

QUESTION ID
S62-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE One full turn is $2\pi \approx 6.28$ radians.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q3/5

A point on the unit circle travels an arc of length π. The angle is

A. 1 radian

B. 2π radians

C. π°

D. π radians (180°)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q4/5

2481

QUESTION ID

S62-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. π radians (180°)

GRADING NOTE θ = s/r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F2-Q4**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the radian measure of an angle does not depend on the radius of the circle used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q5/5

2482

QUESTION ID
S62-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Arc length is proportional to radius, so s/r is the same for every circle.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 62 • FORWARD 2 • Q5/5

SET 63 • HSG-C.B.5 • CIRCLES

Derive and use the formula for the area of a sector

MAIN	Geometry main question	**HSG-C.B.5**
BACKWARD 1	Area of a circle	**7.G.B.4**
BACKWARD 2	The radius is half the diameter	**7.G.B.4**
BACKWARD 3	Fractions of a circle in degrees and radians	**HSG-C.B.5**
FORWARD 1	Solve inverse sector problems	**HSG-C.B.5**
FORWARD 2	Areas of segments and composite circular regions	**HSG-MG.A.1**

SET 63 • HSG-C.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Circles • HSG-C.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Derive a formula for the area of a sector with central angle θ (in degrees) in a circle of radius r. Then rewrite it for θ in radians.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q1/5

2485

QUESTION ID

S63-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

A = (θ/360)πr² for degrees; A = ½r²θ for radians.

GRADING NOTE REQUIRED WORK: Use the fraction of the whole circle that the sector is, times the circle's area.

Any valid proof (paragraph, two-column, or flow). Essential: the sector is θ/360 of the circle (central angles split the area in proportion); area = (θ/360)πr²; with θ radians, the fraction is θ/(2π), so A = (θ/(2π))πr² = ½r²θ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q1/5

2485

ANSWER KEY Reduced copy of the question page, with its answer

Derive the formula A = ½rs for the area of a sector in terms of its radius r and arc length s. Then use it for a sector with r = 8 and s = 5.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q2/5

2486

QUESTION ID

S63-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

A = ½rs; A = 20

GRADING NOTE

REQUIRED WORK: Use the fraction s/(2πr) of the circumference, which is the same fraction of the area.

Any valid proof (paragraph, two-column, or flow). Essential: the sector is s/(2πr) of the circle (arc length and area are in the same proportion to the whole circle); A = (s/(2πr)) · πr² = ½rs; ½ · 8 · 5 = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q2/5

2486

ANSWER KEY Reduced copy of the question page, with its answer

Derive the sector area formula for θ in radians from the proportion $A/(\pi r^2) = \theta/(2\pi)$. Then find the area of a sector with central angle 1.2 radians in a circle of radius 5.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q3/5

2487

QUESTION ID

S63-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

A = ½r²θ; area 15

GRADING NOTE

REQUIRED WORK: Solve the proportion for A, then substitute.

Any valid proof (paragraph, two-column, or flow). Essential: the fraction of the full angle (2π) equals the fraction of the area; $A = \pi r^2 \cdot \theta/(2\pi) = \frac{1}{2}r^2\theta$; $\frac{1}{2} \cdot 25 \cdot 1.2 = 15$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A circular pizza of diameter d is cut into n equal slices. Derive a formula for the area of one slice in terms of d and n, then find the area for d = 14 in. and n = 8, to the nearest tenth of a square inch.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q4/5

2488

QUESTION ID

S63-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

A = πd²/(4n); 19.2 in²

GRADING NOTE REQUIRED WORK: Each slice is a sector with central angle 360°/n and radius d/2.

Any valid proof (paragraph, two-column, or flow). Essential: radius d/2 (not d); fraction 1/n; $A = (1/n)\pi(d/2)^2 = \pi d^2/(4n)$; $\pi \cdot 196/32 = 19.242$. Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve the degree sector formula $A = (\theta/360)\pi r^2$ for θ, and use your formula to find the central angle of a sector with area 12π in a circle of radius 6.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q5/5

2489

QUESTION ID

S63-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

$\theta = 360A/(\pi r^2)$;

$\theta = 120°$

GRADING NOTE REQUIRED WORK: Isolate θ algebraically, then substitute.

Any valid proof (paragraph, two-column, or flow). Essential: multiply both sides by 360 and divide by πr^2; $\theta = 360 \cdot 12\pi/(36\pi) = 120°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • MAIN • Q5/5

SET 63 • HSG-C.B.5

BACKWARD 1

Area of a circle • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the exact area of a circle with radius 6.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q1/5

2491

QUESTION ID

S63-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has diameter 10. Exact area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q2/5

2492

QUESTION ID

S63-B1-Q2

TYPE

Short answer

CORRECT ANSWER

25π

GRADING NOTE Form required: exact value. Use the radius 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$A = \pi r^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q3/5

2493

QUESTION ID

S63-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has area 49π. Radius?

A. 14

B. 7

C. 24.5

D. 49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q4/5

2494

QUESTION ID

S63-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 7

GRADING NOTE $r^2 = 49$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q4/5

True or false?

Doubling the radius doubles the area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q5/5

2495

QUESTION ID

S63-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It multiplies the area by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 1 • Q5/5

SET 63 • HSG-C.B.5

BACKWARD 2

The radius is half the diameter • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A circle has diameter 14 in. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q1/5

2497

QUESTION ID

S63-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7 in.

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q1/5

True or false?

Using the diameter in $A = \pi r^2$ gives an area four times too large.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q2/5

2498

QUESTION ID

S63-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(2r)^2 = 4r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A sector of a circle with diameter 20 uses r =

A. 10

B. 20

C. 5

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q3/5

2499

QUESTION ID

S63-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 10

GRADING NOTE Half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle's radius is 2.5 cm. Diameter?

Diameter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q4/5

2500

QUESTION ID

S63-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q4/5

2500

True or false?

The radius and diameter can be used interchangeably in area formulas.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q5/5

2501

QUESTION ID

S63-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They differ by a factor of 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 63 • BACKWARD 2 • Q5/5

SET 63 • HSG-C.B.5

BACKWARD 3

Fractions of a circle in degrees and radians • High School — Geometry • Geometry: Circles • HSG-C.B.5

ANSWER KEY Reduced copy of the question page, with its answer

What fraction of a circle is a sector with central angle 45°?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q1/5

2503

QUESTION ID

S63-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{8}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q1/5

2503

ANSWER KEY Reduced copy of the question page, with its answer

What fraction of a circle is a sector with central angle π/2 radians?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q2/5

2504

QUESTION ID

S63-B3-Q2

TYPE

Short answer

CORRECT ANSWER

¼

GRADING NOTE Any equivalent form. (π/2)/(2π).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q2/5

A full circle measures

A. 360° or 2π radians

B. 180° or π radians

C. 360 radians

D. 2π°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q3/5

2505

QUESTION ID

S63-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 360° or 2π radians

GRADING NOTE Both units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q3/5

True or false?

A central angle of 1 radian is 1/(2π) of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q4/5

2506

QUESTION ID

S63-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Fraction of 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q4/5

True or false?

A 90° sector is ⅓ of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q5/5

2507

QUESTION ID

S63-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE ¼.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • BACKWARD 3 • Q5/5

SET 63 • HSG-C.B.5

FORWARD 1

Solve inverse sector problems • High School — Geometry • Geometry: Circles • HSG-C.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A sector with a 40° angle has area 4π. Find the radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q1/5

2509

QUESTION ID

S63-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. $(40/360)\pi r^2 = 4\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q1/5

2509

A sector of a circle with radius 10 has area 30. Find the central angle in radians.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q2/5

2510

QUESTION ID

S63-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.6 radians

GRADING NOTE Any equivalent form. ½(100)θ = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If the radius is tripled and the angle stays the same, the sector area is tripled.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q3/5

2511

QUESTION ID

S63-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Area depends on r^2, so it is multiplied by 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A sector has area 50 and arc length 10. Radius?

A. 20

B. 5

C. 2.5

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q4/5

2512

QUESTION ID

S63-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 10

GRADING NOTE A = ½rs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q4/5

Show that the area of a sector is A = ½rs, where s is the arc length.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q5/5

2513

QUESTION ID

S63-F1-Q5

TYPE

Short answer

CORRECT ANSWER

A = ½r²θ and s = rθ, so A = ½r(rθ) = ½rs.

GRADING NOTE Any equivalent derivation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.B.5 • Set 63 • FORWARD 1 • Q5/5

SET 63 • HSG-C.B.5

FORWARD 2

Areas of segments and composite circular regions • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1

Find the exact area of the region between a 90° arc and its chord in a circle of radius 4 (a circular segment).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q1/5

2515

QUESTION ID

S63-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4π − 8

GRADING NOTE Form required: exact value. Sector 4π minus right triangle 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A sprinkler waters a 120° sector of radius 9 m. Find the watered area to the nearest square meter.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q2/5

2516

QUESTION ID
S63-F2-Q2

TYPE
Short answer

CORRECT ANSWER
85 m²

GRADING NOTE Rounding to the nearest whole number required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q2/5

True or false?

The area of a circular segment (smaller than a semicircle) equals the sector area plus the area of the triangle formed by the two radii and the chord.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q3/5

2517

QUESTION ID
S63-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Subtract the triangle: segment = sector − triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An annulus (ring) has inner radius 3 and outer radius 5. Its area is

A. 8π

B. 4π

C. 16π

D. 25π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q4/5

2518

QUESTION ID
S63-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. 16π

GRADING NOTE 25π − 9π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A windshield wiper blade covers the part of its arm from 20 cm to 60 cm from the pivot and sweeps through 120°. Find the area it wipes to the nearest square centimeter.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q5/5

2519

QUESTION ID

S63-F2-Q5

TYPE

Short answer

CORRECT ANSWER

3351 cm²

GRADING NOTE Rounding to the nearest whole number required. $\frac{1}{3}\pi(60^2 - 20^2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 63 • FORWARD 2 • Q5/5

2519

SET 64 • HSG-GPE.A.1 • EXPRESSING GEOMETRIC PROPERTIES WITH EQUATIONS

Derive the equation of a circle

MAIN	Geometry main question	**HSG-GPE.A.1**
BACKWARD 1	Distance between two points with the Pythagorean Theorem	**8.G.B.8**
BACKWARD 2	Use the Pythagorean Theorem with expressions	**8.G.B.7**
BACKWARD 3	Shifting a graph: reading h and k	**HSF-BF.B.3**
FORWARD 1	Write circle equations from other given information	**HSG-GPE.A.1**
FORWARD 2	Prove or disprove that points lie on a circle	**HSG-GPE.B.4**

SET 64 • HSG-GPE.A.1

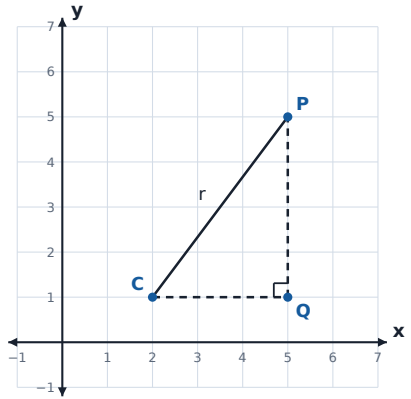
MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A circle has center (h, k) and radius r. Use the Pythagorean Theorem to derive the equation satisfied by every point (x, y) on the circle. The diagram shows one example, with center C(2, 1).

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q1/5

2522

QUESTION ID

S64-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$(x - h)^2 + (y - k)^2 = r^2$

GRADING NOTE REQUIRED WORK: Draw the right triangle with legs $|x - h|$ and $|y - k|$ and hypotenuse r.

Any valid proof (paragraph, two-column, or flow). Essential: a right triangle with horizontal leg $|x - h|$, vertical leg $|y - k|$, hypotenuse r (the distance CP); the Pythagorean Theorem gives $(x - h)^2 + (y - k)^2 = r^2$; squaring removes the absolute values; the reasoning holds for every point on the circle (degenerate legs of length 0 included). Invalid: only checking the one example point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the Pythagorean Theorem (not a memorized formula) to derive the equation of the circle with center (3, −2) and radius 5. Show the right triangle for a general point (x, y) on the circle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q2/5

2523

QUESTION ID

S64-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$(x - 3)^2 + (y + 2)^2 = 25$

GRADING NOTE REQUIRED WORK: Use legs $|x - 3|$ and $|y + 2|$ and hypotenuse 5.

Any valid proof (paragraph, two-column, or flow). Essential: horizontal leg $|x - 3|$, vertical leg $|y - (-2)| = |y + 2|$, hypotenuse 5; $(x - 3)^2 + (y + 2)^2 = 25$ for every point on the circle. Invalid: writing the formula with no triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle is centered at the origin and passes through (6, −8). Use the Pythagorean Theorem (not a memorized formula) to derive its equation, first finding the radius.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q3/5

2524

QUESTION ID

S64-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

$r = 10$; $x^2 + y^2 = 100$

GRADING NOTE REQUIRED WORK: Find the radius as a hypotenuse, then use legs |x| and |y| for a general point.

Any valid proof (paragraph, two-column, or flow). Essential: $r^2 = 6^2 + 8^2 = 100$; for a general point the legs are |x| and |y|, so $x^2 + y^2 = 100$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has center (−1, 4) and passes through (2, 8). Use the Pythagorean Theorem (not a memorized formula) to derive its equation.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q4/5

2525

QUESTION ID

S64-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

$(x + 1)^2 + (y - 4)^2 = 25$

GRADING NOTE REQUIRED WORK: Find r from the right triangle with legs 3 and 4, then write the relationship for a general point (x, y).

Any valid proof (paragraph, two-column, or flow). Essential: legs 3 and 4 give r = 5; general legs |x + 1| and |y − 4|; $(x + 1)^2 + (y - 4)^2 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q4/5

Use the Pythagorean Theorem (not a memorized formula) to derive the equation of the circle with center (−2, 5) and radius 3. Then use your equation to decide whether (−2, 8) is on the circle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q5/5

2526

QUESTION ID

S64-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

$(x + 2)^2 + (y - 5)^2 = 9$; (−2, 8) is on it ($0 + 9 = 9$).

GRADING NOTE

REQUIRED WORK: Use legs $|x + 2|$ and $|y - 5|$ and hypotenuse 3; substitute the point.

Any valid proof (paragraph, two-column, or flow). Essential: the right-triangle relationship with signs handled ($x - (-2) = x + 2$); $r^2 = 9$, not 3; the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-M-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • MAIN • Q5/5

SET 64 • HSG-GPE.A.1

BACKWARD 1

Distance between two points with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (1, 2) and (4, 6).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q1/5

2528

QUESTION ID

S64-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. Legs 3 and 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (0, 0) and (5, 12).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q2/5

2529

QUESTION ID

S64-B1-Q2

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q2/5

The legs of the right triangle joining (2, 1) and (7, 13) are

A. 2 and 1

B. 7 and 13

C. 9 and 14

D. 5 and 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q3/5

2530

QUESTION ID

S64-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5 and 12

GRADING NOTE Differences of coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The distance from $(-3, 0)$ to $(3, 0)$ is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q4/5

2531

QUESTION ID

S64-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Horizontal distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q4/5

True or false?

The distance from (0, 0) to (3, 4) is 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q5/5

2532

QUESTION ID

S64-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 5; the legs are not added.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 64 • BACKWARD 1 • Q5/5

SET 64 • HSG-GPE.A.1

BACKWARD 2

Use the Pythagorean Theorem with expressions • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has legs 6 and 8. Find the hypotenuse.

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q1/5

2534

QUESTION ID

S64-B2-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has legs a and b and hypotenuse c.
Write the Pythagorean relationship.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q2/5

2535

QUESTION ID

S64-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$a^2 + b^2 = c^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q2/5

A right triangle has legs $(x - 2)$ and $(y - 1)$ and hypotenuse 4. Which equation is true?

A. $(x - 2)^2 + (y - 1)^2 = 16$

B. $x^2 + y^2 = 16$

C. $(x - 2)^2 + (y - 1)^2 = 4$

D. $(x - 2) + (y - 1) = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q3/5

2536

QUESTION ID

S64-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $(x - 2)^2 + (y - 1)^2 = 16$

GRADING NOTE Square the legs and the hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q3/5

True or false?

In a right triangle, the hypotenuse is opposite the right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q4/5

2537

GRADING NOTE Definition.

QUESTION ID

S64-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$a^2 + b^2 = c^2$ holds for every triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q5/5

2538

QUESTION ID

S64-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only right triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 64 • BACKWARD 2 • Q5/5

SET 64 • HSG-GPE.A.1

BACKWARD 3

Shifting a graph: reading h and k • High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

The graph of $y = (x - 3)^2$ is the graph of $y = x^2$ shifted in which direction and by how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q1

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q1/5

2540

QUESTION ID

S64-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Right 3

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q1

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write x − (−4) without the double sign.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q2

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q2/5

2541

QUESTION ID

S64-B3-Q2

TYPE

Short answer

CORRECT ANSWER

x + 4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q2

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In $(x + 2)^2$, the shift is

A. 2 down

B. 2 right

C. 2 left

D. 2 up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q3

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q3/5

2542

QUESTION ID
S64-B3-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. 2 left

GRADING NOTE $x - (-2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q3

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Replacing x with x − 5 shifts a graph 5 units right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q4

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q4/5

2543

QUESTION ID
S64-B3-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Horizontal translation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q4

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Replacing y with y + 1 shifts a graph 1 unit up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q5

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q5/5

2544

QUESTION ID
S64-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It shifts it 1 unit down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B3-Q5

High School — Algebra I (course placement in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 64 • BACKWARD 3 • Q5/5

SET 64 • HSG-GPE.A.1

FORWARD 1

Write circle equations from other given information • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A diameter of a circle has endpoints (−2, 3) and (6, −3). Write the equation of the circle.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q1/5

2546

QUESTION ID

S64-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 2)^2 + y^2 = 25$

GRADING NOTE Form required: an equation. Center (2, 0) is the midpoint; radius 5 (half of 10).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F1-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has center (4, −3) and is tangent to the x-axis. Write its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q2/5

2547

QUESTION ID
S64-F1-Q2

TYPE
Short answer

CORRECT ANSWER
$(x - 4)^2 + (y + 3)^2 = 9$

GRADING NOTE Form required: an equation. Radius = distance to the x-axis = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q2/5

True or false?

The circle $(x - 1)^2 + (y - 1)^2 = 2$ passes through the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S64-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q3/5

2548

QUESTION ID

S64-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $1 + 1 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S64-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle with center (0, 5) passes through (0, −1). Its radius is

A. 4

B. 36

C. 6

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q4/5

2549

QUESTION ID

S64-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 6

GRADING NOTE Vertical distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle is centered at (3, 3) and is tangent to both axes. Write its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q5/5

2550

QUESTION ID
S64-F1-Q5

TYPE
Short answer

CORRECT ANSWER
$(x - 3)^2 + (y - 3)^2 = 9$

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 64 • FORWARD 1 • Q5/5

SET 64 • HSG-GPE.A.1

FORWARD 2

Prove or disprove that points lie on a circle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Does (−4, 6) lie on the circle with center (2, −2) passing through (8, 6)? Show the check.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q1/5

2552

QUESTION ID

S64-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: both points are 10 from the center (36 + 64 = 100).

GRADING NOTE Form required: the conclusion with the distance check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F2-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is (7, 1) inside, on, or outside the circle $(x - 3)^2 + (y + 2)^2 = 25$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q2/5

2553

QUESTION ID

S64-F2-Q2

TYPE

Short answer

CORRECT ANSWER

On (16 + 9 = 25)

GRADING NOTE Form required: the conclusion with the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F2-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q2/5

True or false?

A point is inside a circle when its distance from the center is greater than the radius.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S64-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q3/5

2554

QUESTION ID

S64-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Greater than the radius means outside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S64-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q3/5

The point (1, 1) is ___ the circle $x^2 + y^2 = 4$.

A. on

B. outside

C. inside

D. the center of

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q4/5

2555

QUESTION ID

S64-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. inside

GRADING NOTE $1 + 1 = 2 < 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q4/5

A circle has the diameter with endpoints (−2, 3) and (6, −3). Does (5, 4) lie on it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q5/5

2556

QUESTION ID

S64-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Yes:

$(5 - 2)^2 + 4^2 = 25$.

GRADING NOTE Form required: the conclusion with the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 64 • FORWARD 2 • Q5/5

Complete the square to find the center and radius of a circle

MAIN	Geometry main question	**HSG-GPE.A.1**
BACKWARD 1	Complete the square in one variable	**HSA-REI.B.4.a**
BACKWARD 2	Square a binomial	**HSA-APR.A.1**
BACKWARD 3	Keep an equation balanced	**HSA-REI.A.1**
FORWARD 1	Leading coefficients and equations that are not circles	**HSG-GPE.A.1**
FORWARD 2	Complete the square for an ellipse	**HSG-GPE.A.3**

SET 65 • HSG-GPE.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Complete the square to find the center and radius of the circle $x^2 + y^2 - 6x + 4y - 12 = 0$. Show the standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q1/5

2559

QUESTION ID
S65-M-Q1

TYPE
Short answer

CORRECT ANSWER
$(x - 3)^2 + (y + 2)^2 = 25$; center $(3, -2)$; radius 5

GRADING NOTE Form required: both. $(x - 3)^2 + (y + 2)^2 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-M-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Complete the square to find the center and radius of the circle $x^2 + y^2 + 8x - 10y + 5 = 0$. Show the standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q2/5

2560

QUESTION ID
S65-M-Q2

TYPE
Short answer

CORRECT ANSWER
$(x + 4)^2 + (y - 5)^2 = 36$; center $(-4, 5)$; radius 6

GRADING NOTE Form required: both. $(x + 4)^2 + (y - 5)^2 = -5 + 16 + 25 = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-M-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Complete the square to write $x^2 + y^2 - 2x - 14y + 25 = 0$ in standard form. Then give the center and radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q3/5

2561

QUESTION ID

S65-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

$(x - 1)^2 + (y - 7)^2 = 25$; center (1, 7); radius 5

GRADING NOTE REQUIRED WORK: Group x and y terms, add $(b/2)^2$ for each to both sides, and factor.

Form required: the standard form, center, and radius. Added 1 and 49 to both sides: $-25 + 1 + 49 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-M-Q3**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q3/5

Complete the square to find the center and radius of the circle $x^2 + y^2 + 10y = 0$. Show the standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q4/5

2562

QUESTION ID

S65-M-Q4

TYPE

Short answer

CORRECT ANSWER

$x^2 + (y + 5)^2 = 25$; center $(0, -5)$; radius 5

GRADING NOTE Form required: both. $x^2 + (y + 5)^2 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-M-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Complete the square to write $x^2 + y^2 - 4x + 6y = 3$ in standard form. Then give the center and radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q5/5

2563

QUESTION ID

S65-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

$(x - 2)^2 + (y + 3)^2 = 16$; center $(2, -3)$; radius 4

GRADING NOTE REQUIRED WORK: Group x and y terms, add $(b/2)^2$ for each to both sides, and factor.

Form required: the standard form, center, and radius. Added 4 and 9 to both sides: 3 + 4 + 9 = 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • MAIN • Q5/5

SET 65 • HSG-GPE.A.1

BACKWARD 1

Complete the square in one variable • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

What number completes the square: $x^2 - 6x + ___?$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q1/5

2565

QUESTION ID

S65-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form. $(6/2)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B1-Q1**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write $x^2 + 10x + 25$ as a square.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q2/5

2566

QUESTION ID
S65-B1-Q2

TYPE
Short answer

CORRECT ANSWER
$(x + 5)^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B1-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^2 - 8x + 16$ equals

A. $(x + 4)^2$

B. $(x - 8)^2$

C. $(x - 4)^2$

D. $(x - 16)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q3/5

2567

QUESTION ID

S65-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $(x - 4)^2$

GRADING NOTE Half of 8 is 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B1-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

To complete $x^2 + bx$, add $(b/2)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q4/5

2568

QUESTION ID
S65-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x^2 + 6x + 6$ is a perfect square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q5/5

2569

QUESTION ID
S65-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It needs + 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B1-Q5**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 65 • BACKWARD 1 • Q5/5

SET 65 • HSG-GPE.A.1

BACKWARD 2

Square a binomial • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(x - 3)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q1/5

2571

QUESTION ID
S65-B2-Q1

TYPE
Short answer

CORRECT ANSWER
$x^2 - 6x + 9$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(y + 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q2/5

2572

QUESTION ID
S65-B2-Q2

TYPE
Short answer

CORRECT ANSWER
$y^2 + 4y + 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B2-Q2**
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x + 5)^2 =$

A. $x^2 + 10x + 25$

B. $x^2 + 25$

C. $x^2 + 10x + 10$

D. $x^2 + 5x + 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q3/5

2573

QUESTION ID
S65-B2-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. $x^2 + 10x + 25$

GRADING NOTE Middle term 2 · 5x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(a - b)^2 = a^2 - 2ab + b^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q4/5

2574

QUESTION ID
S65-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(x - 3)^2 = x^2 - 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q5/5

2575

QUESTION ID
S65-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Missing the middle term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 65 • BACKWARD 2 • Q5/5

SET 65 • HSG-GPE.A.1

BACKWARD 3

Keep an equation balanced • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

If 9 is added to the left side of an equation, what must be done to the right side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q1/5

2577

QUESTION ID
S65-B3-Q1

TYPE
Short answer

CORRECT ANSWER
Add 9

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^2 - 6x = 7$. Add 9 to both sides and factor.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q2/5

2578

QUESTION ID

S65-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 3)^2 = 16$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-B3-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^2 + y^2 - 4x = 5$ becomes $(x - 2)^2 + y^2 =$

A. 4

B. 5

C. 1

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q3/5

2579

QUESTION ID

S65-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 9

GRADING NOTE 5 + 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q3/5

True or false?

Adding the same number to both sides keeps the solutions the same.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q4/5

2580

QUESTION ID
S65-B3-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Equality property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q4/5

True or false?

A number added to one side need not be added to the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q5/5

2581

QUESTION ID
S65-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE That changes the equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 65 • BACKWARD 3 • Q5/5

SET 65 • HSG-GPE.A.1

FORWARD 1

Leading coefficients and equations that are not circles • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Find the center and radius of
$2x^2 + 2y^2 - 8x + 12y - 6 = 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q1/5

2583

QUESTION ID
S65-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Center (2, −3); radius 4

GRADING NOTE Form required: both. Divide by 2 first.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F1-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q1/5

Does $x^2 + y^2 + 2x + 4y + 10 = 0$ represent a circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q2/5

2584

QUESTION ID

S65-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No: it becomes $(x + 1)^2 + (y + 2)^2 = -5$, and a sum of squares cannot be negative.

GRADING NOTE Form required: the conclusion with the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q2/5

True or false?

$x^2 + y^2 - 6x + 9 = 0$ represents a single point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S65-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q3/5

2585

QUESTION ID

S65-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x - 3)^2 + y^2 = 0$: only (3, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S65-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$3x^2 + 3y^2 - 12y = 0$ has radius

A. $\sqrt{12}$

B. 2

C. 12

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q4/5

2586

QUESTION ID

S65-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 2

GRADING NOTE $x^2 + (y - 2)^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For what values of c is $x^2 + y^2 - 4x + c = 0$ a circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q5/5

2587

QUESTION ID

S65-F1-Q5

TYPE

Short answer

CORRECT ANSWER

c < 4

GRADING NOTE Form required: an inequality. $r^2 = 4 - c > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F1-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 65 • FORWARD 1 • Q5/5

SET 65 • HSG-GPE.A.1

FORWARD 2

Complete the square for an ellipse • High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Write $4x^2 + 9y^2 - 16x + 18y - 11 = 0$ in standard form and give the center.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q1

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q1/5

2589

QUESTION ID

S65-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 2)^2/9 + (y + 1)^2/4 = 1$; center $(2, -1)$

GRADING NOTE Form required: both. $4(x - 2)^2 + 9(y + 1)^2 = 11 + 16 + 9 = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F2-Q1**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the center of $x^2 + 4y^2 + 6x - 8y + 9 = 0$.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q2

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q2/5

2590

QUESTION ID

S65-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(−3, 1)

GRADING NOTE Form required: an ordered pair. $(x + 3)^2 + 4(y - 1)^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F2-Q2**

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q2/5

True or false?

After factoring 4 out of $4x^2 - 16x$, add 16 inside the parentheses to complete the square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q3/5

2591

QUESTION ID

S65-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Add 4 inside: $4(x^2 - 4x + 4)$, which adds 16 to the expression.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is NOT a circle?

A. $x^2 + 4y^2 = 4$

B. $2x^2 + 2y^2 = 8$

C. $x^2 + y^2 = 4$

D. $(x - 1)^2 + y^2 = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q4/5

2592

QUESTION ID

S65-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x^2 + 4y^2 = 4$

GRADING NOTE Unequal coefficients give an ellipse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $x^2 + 4y^2 = 4$ is not a circle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q5/5

2593

QUESTION ID
S65-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The x^2 and y^2 coefficients differ, so the "radius" differs by direction (2 across, 1 up).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School — Geometry, Advanced Extension (+) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3 • Set 65 • FORWARD 2 • Q5/5

Derive the equation of a parabola from a focus and directrix

MAIN	Geometry main question	**HSG-GPE.A.2**
BACKWARD 1	Distance between two points	**8.G.B.8**
BACKWARD 2	Distance from a point to a horizontal line	**6.NS.C.8**
BACKWARD 3	Points equidistant from two objects	**HSG-CO.C.9**
BACKWARD 4	Expand and simplify squares	**HSA-APR.A.1**
FORWARD 1	Find the focus and directrix from an equation	**HSG-GPE.A.2**
FORWARD 2	Design with the reflective property of parabolas	**HSG-MG.A.3**

SET 66 • HSG-GPE.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Derive the equation of the parabola with focus (0, 2) and directrix y = −2.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q1/5

2596

QUESTION ID

S66-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$x^2 = 8y$ ($y = x^2/8$)

GRADING NOTE REQUIRED WORK: Set the distance from (x, y) to the focus equal to its distance to the directrix, square, and simplify.

Any valid proof (paragraph, two-column, or flow). Essential: $\sqrt{(x^2 + (y - 2)^2)} = |y + 2|$; square: $x^2 + y^2 - 4y + 4 = y^2 + 4y + 4$; simplify to $x^2 = 8y$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Derive, from the definition (equal distances to the focus and the directrix), the equation of the parabola with focus (0, −3) and directrix y = 3.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q2/5

2597

QUESTION ID

S66-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$x^2 = -12y$

GRADING NOTE REQUIRED WORK: Set $\sqrt{(x^2 + (y + 3)^2)} = |y - 3|$, square, and simplify.

Any valid proof (paragraph, two-column, or flow). Essential: $x^2 + y^2 + 6y + 9 = y^2 - 6y + 9$; $x^2 = -12y$. Invalid: quoting $x^2 = 4py$ without the distance equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Derive, from the definition (equal distances to the focus and the directrix), the equation of the parabola with focus (2, 5) and directrix y = 1.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q3/5

2598

QUESTION ID

S66-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

$(x - 2)^2 = 8(y - 3)$

GRADING NOTE REQUIRED WORK: Set $\sqrt{((x - 2)^2 + (y - 5)^2)} = |y - 1|$, square, and simplify.

Any valid proof (paragraph, two-column, or flow). Essential: $(x - 2)^2 + y^2 - 10y + 25 = y^2 - 2y + 1$; $(x - 2)^2 = 8y - 24 = 8(y - 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Derive, from the definition (equal distances to the focus and the directrix), the equation of the parabola with focus (3, 0) and directrix $x = -3$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q4/5

2599

QUESTION ID

S66-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

$y^2 = 12x$

GRADING NOTE REQUIRED WORK: Set $\sqrt{((x-3)^2 + y^2)} = |x+3|$, square, and simplify.

Any valid proof (paragraph, two-column, or flow). Essential: $x^2 - 6x + 9 + y^2 = x^2 + 6x + 9$; $y^2 = 12x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Derive, from the definition (equal distances to the focus and the directrix), the equation of the parabola with focus (1, 4) and directrix y = 0 (the x-axis). Then give its vertex.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q5/5

2600

QUESTION ID

S66-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

$(x - 1)^2 = 8(y - 2)$; vertex $(1, 2)$

GRADING NOTE REQUIRED WORK: Set $\sqrt{((x - 1)^2 + (y - 4)^2)} = |y|$, square, simplify, and read the vertex.

Any valid proof (paragraph, two-column, or flow). Essential: $(x - 1)^2 + y^2 - 8y + 16 = y^2$; $(x - 1)^2 = 8y - 16 = 8(y - 2)$; the vertex (1, 2) is midway between the focus and the directrix.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • MAIN • Q5/5

SET 66 • HSG-GPE.A.2

BACKWARD 1

Distance between two points • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance from (3, 6) to (0, 2).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q1/5

2602

QUESTION ID

S66-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the distance from (x, y) to (0, 2) as an expression.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q2/5

2603

QUESTION ID

S66-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$\sqrt{(x^2 + (y - 2)^2)}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The distance from (4, 1) to (1, 5) is

A. 7

B. 5

C. √7

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q3/5

2604

QUESTION ID

S66-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5

GRADING NOTE Legs 3 and 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q3/5

True or false?

The distance formula comes from the Pythagorean Theorem.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q4/5

2605

QUESTION ID

S66-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Legs are coordinate differences.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{(9 + 16)} = 3 + 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q5/5

2606

QUESTION ID

S66-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{25} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 66 • BACKWARD 1 • Q5/5

SET 66 • HSG-GPE.A.2

BACKWARD 2

Distance from a point to a horizontal line • Grade 6 • The Number System • 6.NS.C.8

ANSWER KEY Reduced copy of the question page, with its answer

How far is (5, 7) from the line y = 2?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q1/5

2608

QUESTION ID

S66-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How far is (1, −3) from the line y = 4?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q2/5

2609

QUESTION ID

S66-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The distance from (x, y) to y = −2 is

A. |x + 2|

B. y

C. |y − 2|

D. |y + 2|

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q3/5

2610

QUESTION ID

S66-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. |y + 2|

GRADING NOTE Vertical difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q3/5

2610

True or false?

The distance from a point to a horizontal line is measured straight up or down.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q4/5

2611

QUESTION ID

S66-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q4/5

True or false?

The distance from (6, 1) to y = 4 is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q5/5

2612

QUESTION ID

S66-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 66 • BACKWARD 2 • Q5/5

SET 66 • HSG-GPE.A.2

BACKWARD 3

Points equidistant from two objects • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

The set of points equidistant from two points is the perpendicular bisector of the segment joining them.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q1/5

2614

QUESTION ID

S66-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A parabola is the set of points equidistant from what two objects?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q2/5

2615

QUESTION ID

S66-B3-Q2

TYPE

Short answer

CORRECT ANSWER

A point (the focus) and a line (the directrix)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q2/5

The vertex of a parabola is

A. at the focus

B. twice as far from the focus

C. halfway between the focus and the directrix

D. on the directrix

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q3/5

2616

QUESTION ID

S66-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. halfway between the focus and the directrix

GRADING NOTE Equidistant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Focus (0, 6), directrix y = 0. Find the vertex.

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q4/5

2617

QUESTION ID

S66-B3-Q4

TYPE

Short answer

CORRECT ANSWER

(0, 3)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q4/5

True or false?

Every point of a parabola is closer to the focus than to the directrix.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q5/5

2618

QUESTION ID

S66-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Equal distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 66 • BACKWARD 3 • Q5/5

SET 66 • HSG-GPE.A.2

BACKWARD 4

Expand and simplify squares • High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Expand $(y - 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q1/5

2620

QUESTION ID
S66-B4-Q1

TYPE
Short answer

CORRECT ANSWER
$y^2 - 4y + 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q1
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(y + 2)^2 - (y - 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q2/5

2621

QUESTION ID
S66-B4-Q2

TYPE
Short answer

CORRECT ANSWER
8y

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q2
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$y^2 + 6y + 9 - (y^2 - 6y + 9) =$

A. 18

B. $2y^2$

C. 0

D. 12y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q3/5

2622

QUESTION ID
S66-B4-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. 12y

GRADING NOTE The squares cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q3
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(y + 3)^2 = y^2 + 6y + 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q4/5

2623

QUESTION ID
S66-B4-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q4
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(y - 2)^2 - y^2 = 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q5/5

2624

QUESTION ID
S66-B4-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $-4y + 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B4-Q5
High School — Algebra I (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 66 • BACKWARD 4 • Q5/5

SET 66 • HSG-GPE.A.2

FORWARD 1

Find the focus and directrix from an equation • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2

Find the focus and directrix of $y = x^2/12$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q1/5

2626

QUESTION ID

S66-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Focus (0, 3); directrix y = −3

GRADING NOTE Form required: both. $x^2 = 12y = 4py$, $p = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q1/5

Find the vertex, focus, and directrix of $(x - 1)^2 = -8(y + 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q2/5

2627

QUESTION ID

S66-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Vertex (1, −2); focus (1, −4); directrix y = 0

GRADING NOTE Form required: all three. p = −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q2/5

True or false?

In $x^2 = 4py$ ($p \neq 0$), the focus is $|p|$ units from the vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q3/5

2628

QUESTION ID

S66-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The focus is (0, p); its distance from the vertex (0, 0) is |p|, since p may be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The parabola $y^2 = -20x$ opens

A. left

B. up

C. down

D. right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q4/5

2629

QUESTION ID

S66-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. left

GRADING NOTE Negative coefficient with y^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the focus of $y = 2x^2$.

Focus: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q5/5

2630

QUESTION ID

S66-F1-Q5

TYPE

Short answer

CORRECT ANSWER

(0, ⅛)

GRADING NOTE Form required: an ordered pair. $x^2 = ½y$, $4p = ½$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2 • Set 66 • FORWARD 1 • Q5/5

SET 66 • HSG-GPE.A.2

FORWARD 2

Design with the reflective property of parabolas • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A satellite dish is 4 ft across and 1 ft deep. How far from the vertex should the receiver (at the focus) be placed?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q1/5

2632

QUESTION ID

S66-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1 ft

GRADING NOTE Any equivalent form. $x^2 = 4py$ through (2, 1): $4 = 4p$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A headlight reflector has equation $x^2 = 12y$ (inches). Where should the bulb go?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q2/5

2633

QUESTION ID
S66-F2-Q2

TYPE
Short answer

CORRECT ANSWER
3 in. above the vertex, at (0, 3)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q2/5

True or false?

Rays parallel to the axis of a parabolic mirror reflect through the vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q3/5

2634

QUESTION ID
S66-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE They reflect through the focus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q3/5

A dish has focus 2 m above its vertex. Its equation (vertex at the origin) is

A. $x^2 = 4y$

B. $x^2 = 8y$

C. $y = 8x^2$

D. $x^2 = 2y$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q4/5

2635

QUESTION ID

S66-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $x^2 = 8y$

GRADING NOTE $4p = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A dish has equation $x^2 = 16y$ (meters) and is 8 m across. How deep is it?

Depth: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q5/5

2636

QUESTION ID
S66-F2-Q5

TYPE
Short answer

CORRECT ANSWER
1 m

GRADING NOTE Any equivalent form. 16 = 16y at x = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 66 • FORWARD 2 • Q5/5

Prove or disprove the type of a quadrilateral with coordinates

MAIN	Geometry main question	**HSG-GPE.B.4**
BACKWARD 1	Find the slope between two points	**8.F.B.4**
BACKWARD 2	Find the distance between two points	**8.G.B.8**
BACKWARD 3	Properties that define special quadrilaterals	**HSG-CO.C.11**
FORWARD 1	Coordinate proofs with variables	**HSG-GPE.B.4**
FORWARD 2	Use the classification to find area	**HSG-GPE.B.7**

SET 67 • HSG-GPE.B.4

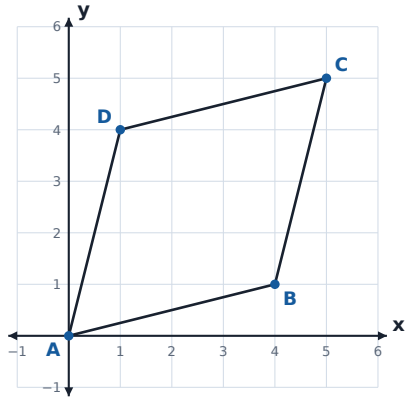
MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Prove that ABCD with A(0, 0), B(4, 1), C(5, 5), D(1, 4) is a rhombus.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q1/5

2639

QUESTION ID

S67-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

AB = BC = CD = DA = √17, so ABCD is a rhombus.

GRADING NOTE REQUIRED WORK: Compute all four side lengths with the distance formula.

Any valid coordinate proof. Essential: all four distances computed correctly (√17 each) and the definition of a rhombus cited. Alternative: show it is a parallelogram (slopes) with perpendicular diagonals (slopes 1 and −1). Invalid: "it looks like a rhombus".

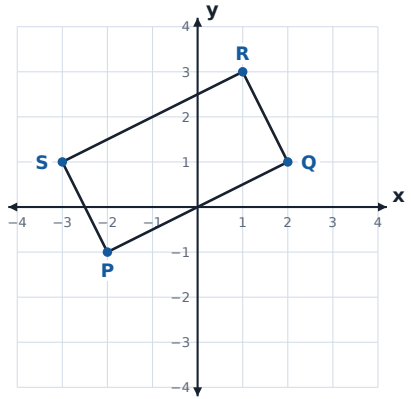
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove that PQRS with P(−2, −1), Q(2, 1), R(1, 3), S(−3, 1) is a rectangle. Is it a square?

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q2/5

2640

QUESTION ID

S67-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Slopes PQ = RS = ½ and QR = SP = −2; adjacent products are −1, so all angles are right: a rectangle. Not a square: PQ = √20 but QR = √5.

GRADING NOTE REQUIRED WORK: Compute the slopes of all four sides; compare adjacent slopes.

Any valid coordinate proof. Essential: four slopes; opposite sides parallel; adjacent slopes multiply to −1 (right angles); side lengths √20 ≠ √5 for "not a square".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove or disprove: KLMN with K(0, 0), L(5, 0), M(6, 3), N(0, 3) is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q3/5

2641

QUESTION ID

S67-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

Disproved: KL ∥ MN (slope 0), but LM has slope 3 while NK is vertical, so LM ∦ NK. KLMN is a trapezoid, not a parallelogram.

GRADING NOTE

REQUIRED WORK: Compare the slopes of both pairs of opposite sides.

Any valid coordinate proof. Essential: one pair of opposite sides shown NOT parallel (a single counterexample disproves). Alternative: KL = 5 but MN = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove that the triangle with vertices A(1, 1), B(4, 5), C(8, 2) is an isosceles right triangle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q4/5

2642

QUESTION ID

S67-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

AB = BC = 5 and AC = √50; $5^2 + 5^2 = 50$, so ∠B is right. Isosceles right triangle.

GRADING NOTE REQUIRED WORK: Use distances for the equal sides and slopes (or the converse of the Pythagorean Theorem) for the right angle.

Any valid coordinate proof. Essential: three distances; two equal (isosceles); right angle at B by the converse of the Pythagorean Theorem or slopes 4/3 and −3/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove or disprove: EFGH with E(−1, 0), F(2, −1), G(3, 2), H(0, 3) is a square.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q5/5

2643

QUESTION ID

S67-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

Proved: all sides are √10; slopes EF = −⅓ and FG = 3 multiply to −1, so ∠F is right. A rhombus with a right angle is a square.

GRADING NOTE REQUIRED WORK: Show four equal sides and one right angle (or a rhombus with a right angle).

Any valid coordinate proof. Essential: four lengths √10; opposite slopes equal (−⅓ and 3); a right angle; conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • MAIN • Q5/5

SET 67 • HSG-GPE.B.4

BACKWARD 1

Find the slope between two points • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope of the line through (1, 2) and (5, 4).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q1/5

2645

QUESTION ID

S67-B1-Q1

TYPE

Short answer

CORRECT ANSWER

½

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope of the line through (0, 3) and (2, −1).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q2/5

2646

QUESTION ID

S67-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The slope through (2, 5) and (6, 5) is

A. undefined

B. 0

C. 5

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q3/5

2647

QUESTION ID

S67-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0

GRADING NOTE Horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q3/5

True or false?

A vertical segment has an undefined slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q4/5

2648

QUESTION ID

S67-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Run 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q4/5

True or false?

Slope is run divided by rise.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q5/5

2649

QUESTION ID

S67-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Rise ÷ run.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 67 • BACKWARD 1 • Q5/5

SET 67 • HSG-GPE.B.4

BACKWARD 2

Find the distance between two points • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (0, 0) and (4, 1).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q1/5

2651

QUESTION ID

S67-B2-Q1

TYPE

Short answer

CORRECT ANSWER

√17

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (1, 1) and (4, 5).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q2/5

2652

QUESTION ID

S67-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which pair of points is 10 apart?

A. (2, 3) and (8, 3)

B. (0, 0) and (6, 8)

C. (0, 0) and (10, 10)

D. (1, 1) and (5, 7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q3/5

2653

GRADING NOTE 36 + 64 = 100.

QUESTION ID

S67-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (0, 0) and (6, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q3/5

True or false?

Two segments with the same length are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q4/5

2654

QUESTION ID

S67-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{10} + \sqrt{10} = \sqrt{20}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q5/5

2655

QUESTION ID

S67-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{10} + \sqrt{10} = 2\sqrt{10} = \sqrt{40}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 67 • BACKWARD 2 • Q5/5

SET 67 • HSG-GPE.B.4

BACKWARD 3

Properties that define special quadrilaterals • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

Which quadrilateral has four congruent sides?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q1/5

2657

QUESTION ID

S67-B3-Q1

TYPE

Short answer

CORRECT ANSWER

A rhombus (a square is a special rhombus)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A parallelogram with one right angle is a ___.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q2/5

2658

QUESTION ID

S67-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Rectangle

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which shows a quadrilateral is a parallelogram?

A. One pair of sides is parallel

B. Two adjacent sides are equal

C. One right angle

D. Both pairs of opposite sides are parallel

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q3/5

2659

QUESTION ID

S67-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Both pairs of opposite sides are parallel

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Every square is a rhombus.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q4/5

2660

QUESTION ID

S67-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Four congruent sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Every rhombus is a square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q5/5

2661

QUESTION ID

S67-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its angles need not be right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 67 • BACKWARD 3 • Q5/5

SET 67 • HSG-GPE.B.4

FORWARD 1

Coordinate proofs with variables • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

A rectangle has vertices O(0, 0), A(a, 0), B(a, b), C(0, b). Prove that its diagonals are congruent.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q1/5

2663

QUESTION ID

S67-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

$OB = AC = \sqrt{(a^2 + b^2)}$

GRADING NOTE

REQUIRED WORK: Compute OB and AC with the distance formula.

Any valid coordinate proof. Essential: $OB = \sqrt{(a^2 + b^2)}$, $AC = \sqrt{((0 - a)^2 + (b - 0)^2)} = \sqrt{(a^2 + b^2)}$; equal for all a, b. Invalid: using numbers for a and b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-F1-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle with vertices O(0, 0), A(2a, 0), B(2b, 2c), where a > 0 and c ≠ 0 (so the points are not collinear). M and N are the midpoints of OB and AB. Prove that MN ∥ OA and MN = ½OA.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q2/5

2664

QUESTION ID

S67-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

M(b, c), N(a + b, c); MN is horizontal like OA, and MN = a = ½(2a).

GRADING NOTE REQUIRED WORK: Find M and N, then compare slope and length.

Any valid coordinate proof. Essential: correct midpoints; slope 0 for both; length a vs 2a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q2/5

True or false?

Placing a figure with a vertex at the origin and a side on the x-axis is allowed in a coordinate proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q3/5

2665

QUESTION ID

S67-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Any figure can be moved there by a rigid motion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q3/5

Why do coordinates like (2a, 0) and (2b, 2c) make the midpoint proof easier?

A. The midpoints have no fractions

B. They make the triangle isosceles

C. They make a right angle

D. They make the slopes 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q4/5

2666

QUESTION ID

S67-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. The midpoints have no fractions

GRADING NOTE Halves of even coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why is a proof using A(6, 0) and B(2, 4) not enough to prove the midsegment theorem?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q5/5

2667

QUESTION ID
S67-F1-Q5

TYPE
Short answer

CORRECT ANSWER
It shows only one triangle; the theorem is about all triangles.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 67 • FORWARD 1 • Q5/5

SET 67 • HSG-GPE.B.4

FORWARD 2

Use the classification to find area • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

ANSWER KEY Reduced copy of the question page, with its answer

ABCD with A(0, 0), B(4, 1), C(5, 5), D(1, 4) is a rhombus. Find its area using its diagonals.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q1/5

2669

QUESTION ID

S67-F2-Q1

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. AC = 5√2, BD = 3√2; ½ · 5√2 · 3√2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rectangle PQRS has P(−2, −1), Q(2, 1), R(1, 3), S(−3, 1). Find its area.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q2/5

2670

QUESTION ID
S67-F2-Q2

TYPE
Short answer

CORRECT ANSWER
10

GRADING NOTE Any equivalent form. $\sqrt{20} \cdot \sqrt{5}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-F2-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q2/5

True or false?

Once a quadrilateral is proved a rhombus, its area is the product of two adjacent side lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q3/5

2671

QUESTION ID

S67-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only for a square; use $\frac{1}{2}d_1d_2$ or base × height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A square has vertices (0, 0), (3, 1), (2, 4), (−1, 3). Its area is

A. 16

B. √10

C. 10

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q4/5

2672

QUESTION ID

S67-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 10

GRADING NOTE Side √10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why proving the shape first matters before using a special area formula.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q5/5

2673

QUESTION ID
S67-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The formula (such as ½d_1d_2 or length × width) is valid only for that type of figure.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 67 • FORWARD 2 • Q5/5

Prove or disprove that a point lies on a circle, and other coordinate proofs

MAIN	Geometry main question	**HSG-GPE.B.4**
BACKWARD 1	Distance between two points	**8.G.B.8**
BACKWARD 2	The equation of a circle	**HSG-GPE.A.1**
BACKWARD 3	Substitute values into an expression	**6.EE.A.2**
FORWARD 1	Coordinate proofs about circles with variables	**HSG-GPE.B.4**
FORWARD 2	Find the tangent line to a circle at a point	**HSG-GPE.B.5**

SET 68 • HSG-GPE.B.4

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A circle is centered at (2, −1) and contains the point (2, 4). Prove or disprove that the point (5, 3) lies on the circle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q1/5

2676

QUESTION ID

S68-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Proved: the radius is 5 and the distance from (2, −1) to (5, 3) is √(9 + 16) = 5.

GRADING NOTE REQUIRED WORK: Find the radius, then compare the distance from the center to (5, 3).

Any valid coordinate proof. Essential: radius $|4 - (-1)| = 5$; distance $\sqrt{(3^2 + 4^2)} = 5$; conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q1/5

Prove or disprove that (5, 2) lies on the circle with center (1, −1) and radius 5.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q2/5

2677

QUESTION ID

S68-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Proved:

$(5 - 1)^2 + (2 + 1)^2 = 16 + 9 = 25 = 5^2$.

GRADING NOTE

REQUIRED WORK: Compare the distance from the center to the point with the radius.

Any valid coordinate proof. Essential: the squared distance 25 computed from the coordinates; compared with $r^2 = 25$; conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q2/5

Prove or disprove that $(-3, 6)$ lies on the circle $(x - 1)^2 + (y - 2)^2 = 30$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q3/5

2678

QUESTION ID

S68-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

Disproved: 16 + 16 = 32 ≠ 30 (the point is outside).

GRADING NOTE REQUIRED WORK: Substitute the point into the equation, or compare its distance from the center with $\sqrt{30}$.

Any valid coordinate proof. Essential: substitution giving 32; 32 ≠ 30; conclusion (outside, since 32 > 30).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has A(0, 0), B(8, 0), C(0, 6). Prove that the midpoint M of hypotenuse BC is equidistant from A, B, and C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q4/5

2679

QUESTION ID

S68-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

M(4, 3);

MA = MB = MC = 5

GRADING NOTE REQUIRED WORK: Find M, then compute MA, MB, MC.

Any valid coordinate proof. Essential: midpoint (4, 3); three distances of 5. Consequence: A, B, C lie on a circle centered at M.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-M-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove that the point (4, −3) lies outside the circle $x^2 + y^2 = 20$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q5/5

2680

QUESTION ID

S68-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

Distance 5 > √20 ≈ 4.47, so the point is outside.

GRADING NOTE REQUIRED WORK: Compare the distance from the center to the point with the radius.

Any valid coordinate proof. Essential: center (0, 0), radius √20; distance √(16 + 9) = 5; 5 > √20 (25 > 20); conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • MAIN • Q5/5

SET 68 • HSG-GPE.B.4

BACKWARD 1

Distance between two points • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (1, −1) to (5, 2)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q1/5

2682

QUESTION ID

S68-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (2, −1) to (5, 3)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q2/5

2683

QUESTION ID

S68-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q2/5

Distance from (0, 0) to (−3, 4) is

A. 5

B. 1

C. 7

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q3/5

2684

QUESTION ID

S68-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

GRADING NOTE 9 + 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q3/5

True or false?

$(-3)^2 = 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q4/5

2685

QUESTION ID

S68-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Square of a negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q4/5

True or false?

The distance from (2, 2) to (2, 7) is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q5/5

2686

QUESTION ID

S68-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 68 • BACKWARD 1 • Q5/5

SET 68 • HSG-GPE.B.4

BACKWARD 2

The equation of a circle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the circle with center (1, −1) and radius 5.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q1/5

2688

QUESTION ID

S68-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 1)^2 + (y + 1)^2 = 25$

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the radius of $(x - 1)^2 + (y - 2)^2 = 30$?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q2/5

2689

QUESTION ID

S68-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$\sqrt{30}$

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-B2-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Center of $(x + 4)^2 + y^2 = 9$?

A. (4, 0)

B. (−4, 0)

C. (−4, 9)

D. (0, −4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q3/5

2690

QUESTION ID

S68-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (−4, 0)

GRADING NOTE Opposite sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q3/5

True or false?

A point satisfies a circle's equation exactly when it is on the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q4/5

2691

QUESTION ID

S68-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Meaning of the equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q4/5

True or false?

The radius of $x^2 + y^2 = 20$ is 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q5/5

2692

QUESTION ID

S68-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{20}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 68 • BACKWARD 2 • Q5/5

SET 68 • HSG-GPE.B.4

BACKWARD 3

Substitute values into an expression • Grade 6 • Expressions & Equations • 6.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $(x - 1)^2 + (y - 2)^2$ when $x = -3$ and $y = 6$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q1/5

2694

QUESTION ID

S68-B3-Q1

TYPE

Short answer

CORRECT ANSWER

32

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $x^2 + y^2$ when $x = 4$, $y = -3$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q2/5

2695

QUESTION ID

S68-B3-Q2

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x + 1)^2$ when $x = -4$ is

A. -9

B. 9

C. 25

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q3/5

2696

QUESTION ID

S68-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 9

GRADING NOTE $(-3)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q3/5

True or false?

$(-5 - 1)^2 = 36$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q4/5

2697

QUESTION ID

S68-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(-6)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(2 + 1)^2 = 2^2 + 1^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q5/5

2698

QUESTION ID

S68-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $9 \neq 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 68 • BACKWARD 3 • Q5/5

SET 68 • HSG-GPE.B.4

FORWARD 1

Coordinate proofs about circles with variables • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A(−r, 0) and B(r, 0) are endpoints of a diameter of the circle $x^2 + y^2 = r^2$, and P(x, y) is any other point on the circle. Prove that ∠APB is a right angle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q1/5

2700

QUESTION ID

S68-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

slope PA · slope PB = $y^2/(x^2 - r^2)$ = $y^2/(-y^2) = -1$, so PA ⊥ PB.

GRADING NOTE REQUIRED WORK: Compute the slopes of PA and PB and use $x^2 + y^2 = r^2$.

Any valid coordinate proof. Essential: slopes $y/(x + r)$ and $y/(x - r)$; product $y^2/(x^2 - r^2)$; substitute $x^2 - r^2 = -y^2$; product −1. (P is not A or B, so $y \neq 0$.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Show that if (a, b) is on $x^2 + y^2 = r^2$, then (b, a) and (−a, b) are too.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q2/5

2701

QUESTION ID

S68-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$b^2 + a^2 = r^2$ and $(-a)^2 + b^2 = r^2$, the same sum.

GRADING NOTE Any equivalent argument.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F1-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q2/5

True or false?

A coordinate proof with variables covers every case at once.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q3/5

2702

QUESTION ID
S68-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE General.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

To prove four points lie on one circle, show they are

A. all the same distance from one point

B. the same distance from the origin only

C. vertices of a rectangle only

D. all on one line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q4/5

2703

QUESTION ID

S68-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. all the same distance from one point

GRADING NOTE Common center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove or disprove: A(1, 0), B(0, 1), C(−1, 0), D(0, −2) lie on one circle centered at the origin.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q5/5

2704

QUESTION ID

S68-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Disproved: D is 2 from the origin, the others 1.

GRADING NOTE Form required: the conclusion with distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 68 • FORWARD 1 • Q5/5

SET 68 • HSG-GPE.B.4

FORWARD 2

Find the tangent line to a circle at a point • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Find the equation of the line tangent to $x^2 + y^2 = 25$ at (3, 4).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q1/5

2706

QUESTION ID

S68-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$y = -\frac{3}{4}x + 25/4$

GRADING NOTE Form required: an equation; equivalent forms (3x + 4y = 25) accepted. Radius slope 4/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F2-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the tangent to $(x - 1)^2 + (y - 1)^2 = 25$ at (4, 5).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q2/5

2707

QUESTION ID

S68-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$y - 5 = -\frac{3}{4}(x - 4)$

GRADING NOTE Form required: an equation; equivalent forms accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F2-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q2/5

True or false?

The tangent at P is parallel to the radius drawn to P.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q3/5

2708

QUESTION ID
S68-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is perpendicular to that radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q3/5

The radius to (0, 5) on $x^2 + y^2 = 25$ is vertical, so the tangent there is

A. $x = 0$

B. $x = 5$

C. $y = x + 5$

D. $y = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q4/5

2709

QUESTION ID

S68-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. y = 5

GRADING NOTE Horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how the slope of the tangent at (a, b) on $x^2 + y^2 = r^2$ is found.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q5/5

2710

QUESTION ID
S68-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The radius has slope b/a, so the tangent has slope −a/b (negative reciprocal), for a, b ≠ 0.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 68 • FORWARD 2 • Q5/5

SET 69 • HSG-GPE.B.5 • EXPRESSING GEOMETRIC PROPERTIES WITH EQUATIONS

Prove the slope criteria for parallel and perpendicular lines

MAIN	Geometry main question	**HSG-GPE.B.5**
BACKWARD 1	Similar slope triangles give the same slope	**8.EE.B.6**
BACKWARD 2	Rotate a point 90° about the origin	**HSG-CO.A.5**
BACKWARD 3	Negative reciprocals	**7.NS.A.2**
FORWARD 1	Use the criteria to classify pairs of lines	**HSG-GPE.B.5**
FORWARD 2	Use the criteria inside a general coordinate proof	**HSG-GPE.B.4**

SET 69 • HSG-GPE.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Prove that two distinct nonvertical lines with the same slope m are parallel.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q1/5

2713

QUESTION ID

S69-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$mx + b_1 = mx + b_2$ gives $b_1 = b_2$, a contradiction; so the lines never meet and are parallel.

GRADING NOTE REQUIRED WORK: Write the lines as $y = mx + b_1$ and $y = mx + b_2$ with $b_1 \neq b_2$ and show there is no intersection.

Any valid proof (paragraph, two-column, or flow). Essential: equations with equal slopes and different intercepts; solving for an intersection leads to $b_1 = b_2$ (false); conclusion. Alternative: a translation maps one line to the other; or congruent slope triangles with corresponding angles equal.

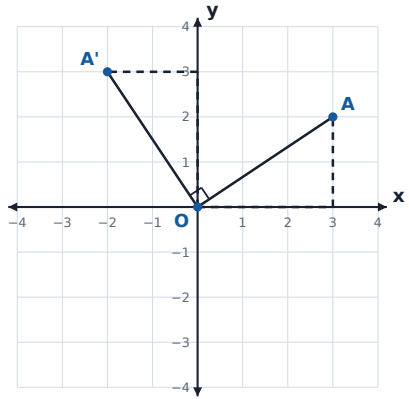
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Line OA passes through O(0, 0) and A(a, b) with a, b ≠ 0. Rotating A 90° counterclockwise about O gives A'(−b, a). Use this to prove that perpendicular lines have slopes whose product is −1. The diagram shows the case A(3, 2).

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q2/5

2714

QUESTION ID

S69-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

slope OA = b/a, slope OA' = a/(−b) = −a/b, product −1.

GRADING NOTE REQUIRED WORK: Find the slopes of OA and OA', then their product.

Any valid proof (paragraph, two-column, or flow). Essential: a 90° rotation maps OA onto a perpendicular line; the image of (a, b) is (−b, a); slopes b/a and −a/b; product −1. Any pair of perpendicular nonvertical lines can be translated to pass through O, keeping slopes. Invalid: checking only the one example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q2/5

Prove that two distinct nonvertical lines that are parallel have equal slopes.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q3/5

2715

QUESTION ID

S69-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

If $m_1 \neq m_2$, then $m_1x + b_1 = m_2x + b_2$ has the solution $x = (b_2 - b_1)/(m_1 - m_2)$, so the lines meet, contradicting parallel. So $m_1 = m_2$.

GRADING NOTE

REQUIRED WORK: Indirect: assume the slopes m_1 and m_2 are different and find an intersection point.

Any valid proof (paragraph, two-column, or flow). Essential: assumption of unequal slopes; solving for the intersection; contradiction with "parallel"; conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove that if two nonvertical lines have slopes m and $-1/m$ ($m \neq 0$), they are perpendicular. (Translate both lines so they pass through the origin first.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q4/5

2716

QUESTION ID

S69-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

(1, m) rotates to (−m, 1), on the line with slope 1/(−m) = −1/m; so the line with slope −1/m through O is the 90° image of y = mx, hence perpendicular.

GRADING NOTE

REQUIRED WORK: Rotate the point (1, m) on y = mx by 90° about the origin and find the slope of its image line.

Any valid proof (paragraph, two-column, or flow). Essential: translations keep slopes; the rotation image of (1, m); its slope −1/m; a 90° rotation image is perpendicular; only one line through O has slope −1/m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Lines $y = mx$ and $y = -x/m$ ($m \neq 0$) pass through the origin O. Use the points P(1, m) and Q(−m, 1) and the converse of the Pythagorean Theorem to prove that the lines are perpendicular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q5/5

2717

QUESTION ID

S69-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

$OP^2 + OQ^2 = (1 + m^2) + (m^2 + 1) = 2 + 2m^2 = PQ^2$, so $\angle POQ = 90°$.

GRADING NOTE REQUIRED WORK: Compute OP^2, OQ^2, and PQ^2, and compare.

Any valid proof (paragraph, two-column, or flow). Essential: $OP^2 = 1 + m^2$; $OQ^2 = m^2 + 1$; $PQ^2 = (1 + m)^2 + (m - 1)^2 = 2 + 2m^2$; converse of the Pythagorean Theorem; conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-M-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • MAIN • Q5/5

SET 69 • HSG-GPE.B.5

BACKWARD 1

Similar slope triangles give the same slope • Grade 8 • Expressions & Equations • 8.EE.B.6

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Any two slope triangles on the same line are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q1/5

2719

QUESTION ID

S69-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A line passes through (0, 0) and (2, 3). Find the rise for a run of 4.

Rise: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q2/5

2720

QUESTION ID

S69-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q2/5

Slope triangles with run 2, rise 3 and run 6, rise 9 give slopes

A. 6 and 18

B. 3/2 and 9/6 that differ

C. both 3/2

D. 2/3 and 6/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q3/5

2721

QUESTION ID

S69-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. both 3/2

GRADING NOTE 9/6 = 3/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the slope of a line is the same between any two of its points.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q4/5

2722

QUESTION ID

S69-B1-Q4

TYPE

Short answer

CORRECT ANSWER

The slope triangles are similar, so rise/run is the same ratio.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The slope between two points on a line depends on which points are chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q5/5

2723

QUESTION ID

S69-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 69 • BACKWARD 1 • Q5/5

SET 69 • HSG-GPE.B.5

BACKWARD 2

Rotate a point 90° about the origin • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

ANSWER KEY Reduced copy of the question page, with its answer

Rotate (3, 2) 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q1/5

2725

QUESTION ID

S69-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(−2, 3)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rotate (1, 4) 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q2/5

2726

QUESTION ID

S69-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(−4, 1)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q2/5

The rule for a 90° counterclockwise rotation about the origin is

A. $(x, y) \to (-y, x)$

B. $(x, y) \to (-x, -y)$

C. $(x, y) \to (y, x)$

D. $(x, y) \to (y, -x)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q3/5

2727

QUESTION ID

S69-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (x, y) → (−y, x)

GRADING NOTE Rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q3/5

True or false?

A 90° rotation maps a line to a perpendicular line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q4/5

2728

QUESTION ID

S69-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Angle of rotation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q4/5

True or false?

A 90° rotation maps (2, 0) to (0, −2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q5/5

2729

QUESTION ID

S69-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Counterclockwise gives (0, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 69 • BACKWARD 2 • Q5/5

SET 69 • HSG-GPE.B.5

BACKWARD 3

Negative reciprocals • Grade 7 • The Number System • 7.NS.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Negative reciprocal of 2/3?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q1/5

2731

QUESTION ID

S69-B3-Q1

TYPE

Short answer

CORRECT ANSWER

−3/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply (b/a)(−a/b).

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q2/5

2732

QUESTION ID

S69-B3-Q2

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Negative reciprocal of −4?

A. ¼

B. −¼

C. 4

D. −4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q3/5

2733

QUESTION ID

S69-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. ¼

GRADING NOTE −1 ÷ (−4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A number times its negative reciprocal is −1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q4/5

2734

QUESTION ID

S69-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The negative reciprocal of 1 is 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q5/5

2735

QUESTION ID

S69-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 69 • BACKWARD 3 • Q5/5

SET 69 • HSG-GPE.B.5

FORWARD 1

Use the criteria to classify pairs of lines • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Are y = 3x − 2 and 6x − 2y = 7 parallel, perpendicular, or neither?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q1/5

2737

QUESTION ID

S69-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Parallel (both slopes 3)

GRADING NOTE Form required: the classification with the slopes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Are 2x + 5y = 1 and 5x − 2y = 4 parallel, perpendicular, or neither?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q2/5

2738

QUESTION ID

S69-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Perpendicular (slopes −2/5 and 5/2)

GRADING NOTE Form required: the classification with the slopes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q2/5

True or false?

A horizontal line and a vertical line are perpendicular even though the slope product rule cannot be used.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q3/5

2739

QUESTION ID

S69-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Vertical slope is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q3/5

$y = \frac{1}{2}x + 1$ and $y = 2x - 1$ are

A. neither

B. the same line

C. perpendicular

D. parallel

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q4/5

2740

QUESTION ID

S69-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. neither

GRADING NOTE Product 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find k so that y = kx + 1 is perpendicular to y = −⅓x.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q5/5

2741

QUESTION ID

S69-F1-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F1-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 69 • FORWARD 1 • Q5/5

SET 69 • HSG-GPE.B.5

FORWARD 2

Use the criteria inside a general coordinate proof • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Rhombus OABC has O(0, 0), A(a, 0), B(a + b, c), C(b, c), with $b^2 + c^2 = a^2$. Prove that its diagonals are perpendicular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q1/5

2743

QUESTION ID

S69-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

slope OB · slope AC = c/(a + b) · c/(b − a) = c²/(b² − a²) = c²/(−c²) = −1

GRADING NOTE REQUIRED WORK: Find the slopes of OB and AC and multiply.

Any valid coordinate proof. Essential: both slopes; product simplified with $b^2 + c^2 = a^2$; conclusion. (Assume $b \neq \pm a$, so neither diagonal is vertical.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q1/5

A coordinate proof places rhombus OABC at O(0, 0), A(a, 0), B(a + b, c), C(b, c). Why must the proof assume $b^2 + c^2 = a^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q2/5

2744

QUESTION ID

S69-F2-Q2

TYPE

Short answer

CORRECT ANSWER

It says OC = a = OA, the equal sides that make OABC a rhombus.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q2/5

True or false?

The slope criterion turns "perpendicular" into an algebra statement.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q3/5

2745

QUESTION ID
S69-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Product −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q3/5

In a coordinate proof that lines are parallel, you show

A. equal slopes

B. equal intercepts

C. slope product −1

D. equal lengths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q4/5

2746

QUESTION ID

S69-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. equal slopes

GRADING NOTE Criterion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q4/5

Prove or disprove with slopes: A(0, 0), B(4, 2), C(3, 4) form a right angle at B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q5/5

2747

QUESTION ID

S69-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Proved: slope AB = ½ and slope BC = −2; the product is −1, so AB ⊥ BC.

GRADING NOTE Form required: the conclusion with slopes. Slope BC = (4 − 2)/(3 − 4) = −2; product −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 69 • FORWARD 2 • Q5/5

SET 70 • HSG-GPE.B.5 • EXPRESSING GEOMETRIC PROPERTIES WITH EQUATIONS

Write equations of parallel and perpendicular lines

MAIN	Geometry main question	**HSG-GPE.B.5**
BACKWARD 1	Read the slope of a line	**8.F.B.4**
BACKWARD 2	Rewrite an equation in slope-intercept form	**HSA-CED.A.4**
BACKWARD 3	Write a line from a point and a slope	**HSF-LE.A.2**
BACKWARD 4	Opposite reciprocals	**7.NS.A.2**
FORWARD 1	Write perpendicular bisectors and altitudes	**HSG-GPE.B.5**
FORWARD 2	Find the distance from a point to a line	**HSG-GPE.B.4**

SET 70 • HSG-GPE.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line parallel to y = 3x − 5 that passes through (2, 4).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q1/5

2750

QUESTION ID
S70-M-Q1

TYPE
Short answer

CORRECT ANSWER
y = 3x − 2

GRADING NOTE Form required: an equation; equivalent forms accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line perpendicular to $y = \frac{2}{3}x + 1$ that passes through $(4, -1)$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q2/5

2751

QUESTION ID

S70-M-Q2

TYPE

Short answer

CORRECT ANSWER

$y = -3/2\ x + 5$

GRADING NOTE Form required: an equation; equivalent forms accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-M-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line parallel to 2x + 5y = 10 that passes through (5, 3).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q3/5

2752

QUESTION ID
S70-M-Q3

TYPE
Short answer

CORRECT ANSWER
y = −2/5 x + 5

GRADING NOTE Form required: an equation; equivalent forms (2x + 5y = 25) accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line perpendicular to x = 4 that passes through (1, −3).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q4/5

2753

QUESTION ID
S70-M-Q4

TYPE
Short answer

CORRECT ANSWER
y = −3

GRADING NOTE Form required: an equation. x = 4 is vertical, so the line is horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q4/5

Which line is perpendicular to y = −4x + 1 and passes through the origin?

A. y = −4x

B. y = −¼x

C. y = ¼x

D. y = 4x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q5/5

2754

QUESTION ID

S70-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. y = ¼x

GRADING NOTE Negative reciprocal of −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • MAIN • Q5/5

SET 70 • HSG-GPE.B.5

BACKWARD 1

Read the slope of a line • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Slope of $y = -2x + 7$?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q1/5

2756

QUESTION ID

S70-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Slope of y = 5?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q2/5

2757

QUESTION ID

S70-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Slope of $y = \frac{2}{3}x - 4$?

A. 3/2

B. ⅔

C. −4

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q3/5

2758

QUESTION ID

S70-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. ⅔

GRADING NOTE Coefficient of x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In y = mx + b, m is the slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q4/5

2759

QUESTION ID

S70-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Slope-intercept form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The slope of x = 3 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q5/5

2760

QUESTION ID

S70-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 70 • BACKWARD 1 • Q5/5

SET 70 • HSG-GPE.B.5

BACKWARD 2

Rewrite an equation in slope-intercept form • High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2x + 5y = 10 for y.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q1/5

2762

QUESTION ID

S70-B2-Q1

TYPE

Short answer

CORRECT ANSWER

y = −2/5 x + 2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B2-Q1**

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 3x − y = 4 for y.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q2/5

2763

QUESTION ID

S70-B2-Q2

TYPE

Short answer

CORRECT ANSWER

y = 3x − 4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B2-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

6x + 3y = 9 in slope-intercept form is

A. y = 2x + 3

B. y = −6x + 9

C. y = −2x + 9

D. y = −2x + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q3/5

2764

QUESTION ID

S70-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = −2x + 3

GRADING NOTE Divide by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B2-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x - 2y = 6$ has slope ½.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q4/5

2765

QUESTION ID

S70-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $y = \frac{1}{2}x - 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

4x + 2y = 8 has slope 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q5/5

2766

QUESTION ID

S70-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 70 • BACKWARD 2 • Q5/5

SET 70 • HSG-GPE.B.5

BACKWARD 3

Write a line from a point and a slope • High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Write the line with slope 3 through (2, 4).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q1

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q1/5

2768

QUESTION ID

S70-B3-Q1

TYPE

Short answer

CORRECT ANSWER

y = 3x − 2

GRADING NOTE Form required: an equation; equivalent forms accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B3-Q1**

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the line with slope −1 through (0, 6).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q2

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q2/5

2769

QUESTION ID

S70-B3-Q2

TYPE

Short answer

CORRECT ANSWER

y = −x + 6

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q2

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Slope 2 through (1, 5): b =

A. 5

B. 7

C. −3

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q3

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q3/5

2770

QUESTION ID

S70-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3

GRADING NOTE 5 = 2 + b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B3-Q3**

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y - 4 = 3(x - 2)$ is the line with slope 3 through (2, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q4

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q4/5

2771

QUESTION ID
S70-B3-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Point-slope form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q4

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A line through (0, 6) has y-intercept 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q5

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q5/5

2772

QUESTION ID

S70-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q5

High School — Algebra I (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 70 • BACKWARD 3 • Q5/5

SET 70 • HSG-GPE.B.5

BACKWARD 4

Opposite reciprocals • Grade 7 • The Number System • 7.NS.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Negative reciprocal of ⅔?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q1/5

2774

QUESTION ID

S70-B4-Q1

TYPE

Short answer

CORRECT ANSWER

−3/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Negative reciprocal of −5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q2/5

2775

QUESTION ID

S70-B4-Q2

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{5}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Negative reciprocal of −2/5?

A. 5/2

B. 2/5

C. −2/5

D. −5/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q3/5

2776

QUESTION ID

S70-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5/2

GRADING NOTE Flip and change sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q3/5

True or false?

The negative reciprocal of ¼ is −4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q4/5

2777

QUESTION ID

S70-B4-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Flip and change sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The negative reciprocal of 3 is −3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q5/5

2778

QUESTION ID

S70-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE −⅓.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B4-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 70 • BACKWARD 4 • Q5/5

SET 70 • HSG-GPE.B.5

FORWARD 1

Write perpendicular bisectors and altitudes • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the perpendicular bisector of the segment from (2, 1) to (6, 5).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q1/5

2780

QUESTION ID

S70-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$y = -x + 7$

GRADING NOTE Form required: an equation. Midpoint (4, 3), slope −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F1-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has A(0, 0), B(6, 0), C(2, 4). Write the equation of the altitude from A to BC.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q2/5

2781

QUESTION ID
S70-F1-Q2

TYPE
Short answer

CORRECT ANSWER
y = x

GRADING NOTE Form required: an equation. BC slope −1, so the altitude has slope 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q2/5

True or false?

A perpendicular bisector passes through the midpoint of the segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q3/5

2782

QUESTION ID

S70-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q3/5

The perpendicular bisector of the segment from (0, 0) to (0, 8) is

A. x = 0

B. y = 4

C. x = 4

D. y = 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q4/5

2783

QUESTION ID

S70-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. y = 4

GRADING NOTE Horizontal through (0, 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the perpendicular bisector of the segment from (−3, 2) to (5, 2).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q5/5

2784

QUESTION ID

S70-F1-Q5

TYPE

Short answer

CORRECT ANSWER

x = 1

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F1-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 70 • FORWARD 1 • Q5/5

SET 70 • HSG-GPE.B.5

FORWARD 2

Find the distance from a point to a line • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance from P(5, 5) to the line y = −x + 2.
Give the exact value.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q1/5

2786

QUESTION ID

S70-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

4√2 (about 5.66); the foot is (1, 1)

GRADING NOTE REQUIRED WORK: Write the perpendicular through P, find where it meets the line, then use the distance formula. Form required: exact value (a decimal to 0.01 is also accepted). Perpendicular y = x meets y = −x + 2 at (1, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F2-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance from (0, 0) to the line y = 4.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q2/5

2787

QUESTION ID

S70-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q2/5

True or false?

The distance from a point to a line can be measured along any segment from the point to the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q3/5

2788

QUESTION ID
S70-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is measured along the perpendicular, the shortest segment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q3/5

The foot of the perpendicular from (3, 0) to y = x is

A. (0, 3)

B. (0, 0)

C. (3, 3)

D. (1.5, 1.5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q4/5

2789

QUESTION ID

S70-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (1.5, 1.5)

GRADING NOTE y = −x + 3 meets y = x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F2-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance from (6, 1) to the line y = 2x − 1. Give the exact value.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q5/5

2790

QUESTION ID

S70-F2-Q5

TYPE

Short answer

CORRECT ANSWER

2√5 (about 4.47)

GRADING NOTE Form required: exact value (a decimal to 0.01 is also accepted). Foot (2, 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F2-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 70 • FORWARD 2 • Q5/5

Solve geometric problems with the slope criteria

MAIN	Geometry main question	**HSG-GPE.B.5**
BACKWARD 1	Slope from two points	**8.F.B.4**
BACKWARD 2	Solve a linear equation for an unknown	**HSA-REI.B.3**
BACKWARD 3	Equations of parallel and perpendicular lines	**HSG-GPE.B.5**
BACKWARD 4	Opposite sides of a parallelogram are parallel	**HSG-CO.C.11**
FORWARD 1	Find the orthocenter of a triangle	**HSG-GPE.B.5**
FORWARD 2	Shortest path to a straight line in a design	**HSG-MG.A.3**

SET 71 • HSG-GPE.B.5

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

The line through (1, 2) and (4, k) is parallel to y = 2x + 7. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q1/5

2793

QUESTION ID

S71-M-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form. (k − 2)/3 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a parallelogram with A(0, 0), B(5, 1), C(7, 4). Find D.

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q2/5

2794

QUESTION ID

S71-M-Q2

TYPE

Short answer

CORRECT ANSWER

(2, 3)

GRADING NOTE Form required: an ordered pair. AD ∥ BC and DC ∥ AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The segment from (2, 1) to (5, k) is perpendicular to the line y = 3x. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q3/5

2795

QUESTION ID
S71-M-Q3

TYPE
Short answer

CORRECT ANSWER
0

GRADING NOTE Any equivalent form. $(k - 1)/3 = -\frac{1}{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has A(0, 0), B(8, 4), C(4, 6). Find the foot of the altitude from C to AB.

Foot: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q4/5

2796

QUESTION ID

S71-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

(5.6, 2.8)

GRADING NOTE REQUIRED WORK: Write line AB and the perpendicular through C, then solve the system.

Form required: an ordered pair. AB: y = ½x; altitude: y = −2x + 14.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q4/5

Triangle ABC has A(1, 1), B(4, 2), C(3, 5). At which vertex is the right angle?

A. C

B. B

C. A

D. There is no right angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q5/5

2797

QUESTION ID

S71-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. B

GRADING NOTE Slope AB = ⅓, slope BC = −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • MAIN • Q5/5

SET 71 • HSG-GPE.B.5

BACKWARD 1

Slope from two points • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Slope through (1, 2) and (4, 8)?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q1/5

2799

QUESTION ID

S71-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Slope through (4, 2) and (3, 5)?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q2/5

2800

QUESTION ID

S71-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q2/5

Slope through (0, 0) and (8, 4)?

A. ½

B. 8

C. 4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q3/5

2801

QUESTION ID

S71-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. ½

GRADING NOTE 4/8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The slope from A to B equals the slope from B to A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q4/5

2802

QUESTION ID

S71-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The slope through (1, 1) and (1, 5) is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q5/5

2803

QUESTION ID

S71-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 71 • BACKWARD 1 • Q5/5

SET 71 • HSG-GPE.B.5

BACKWARD 2

Solve a linear equation for an unknown • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(k - 2)/3 = 2$.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q1/5

2805

QUESTION ID

S71-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-B2-Q1**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve ½x = −2x + 14.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q2/5

2806

QUESTION ID

S71-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5.6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-B2-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(k - 1)/3 = -\frac{1}{3}$.

A. -2

B. 1

C. 2

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q3/5

2807

QUESTION ID

S71-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0

GRADING NOTE $k - 1 = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-B2-Q3**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Multiplying both sides by 3 is a valid first step in $(k - 2)/3 = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q4/5

2808

QUESTION ID
S71-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Inverse operation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The solution of $2x + 1 = 7$ is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q5/5

2809

QUESTION ID

S71-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 71 • BACKWARD 2 • Q5/5

SET 71 • HSG-GPE.B.5

BACKWARD 3

Equations of parallel and perpendicular lines • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Line perpendicular to y = ½x through (4, 6)?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q1/5

2811

QUESTION ID

S71-B3-Q1

TYPE

Short answer

CORRECT ANSWER

y = −2x + 14

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Line parallel to y = −x through (0, 3)?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q2/5

2812

QUESTION ID

S71-B3-Q2

TYPE

Short answer

CORRECT ANSWER

y = −x + 3

GRADING NOTE Form required: an equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Perpendicular slope to ⅓?

A. 3

B. −3

C. −⅓

D. ⅓

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q3/5

2813

QUESTION ID

S71-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −3

GRADING NOTE Negative reciprocal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q3/5

True or false?

An altitude is perpendicular to the side it meets.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q4/5

2814

QUESTION ID

S71-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q4/5

True or false?

An altitude passes through the midpoint of the opposite side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q5/5

2815

QUESTION ID

S71-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That is a median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • BACKWARD 3 • Q5/5

SET 71 • HSG-GPE.B.5

BACKWARD 4

Opposite sides of a parallelogram are parallel • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

True or false?

In a parallelogram, both pairs of opposite sides are parallel and congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q1/5

2817

QUESTION ID

S71-B4-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A parallelogram has A(0, 0), B(3, 0), C(4, 2). Find D (in order ABCD).

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q2/5

2818

QUESTION ID

S71-B4-Q2

TYPE

Short answer

CORRECT ANSWER

(1, 2)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In parallelogram ABCD, side AB is parallel to

A. AC

B. BC

C. AD

D. DC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q3/5

2819

QUESTION ID

S71-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. DC

GRADING NOTE Opposite side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In parallelogram ABCD, going from B to C is "right 2, up 3". Going from A to D is

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q4/5

2820

QUESTION ID

S71-B4-Q4

TYPE

Short answer

CORRECT ANSWER

Right 2, up 3

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A parallelogram's adjacent sides are always perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q5/5

2821

QUESTION ID

S71-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only rectangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B4-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 71 • BACKWARD 4 • Q5/5

SET 71 • HSG-GPE.B.5

FORWARD 1

Find the orthocenter of a triangle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has A(0, 0), B(6, 0), C(2, 4). Find the orthocenter (the intersection of the altitudes).

Orthocenter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q1/5

2823

QUESTION ID

S71-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

(2, 2)

GRADING NOTE REQUIRED WORK: Write two altitude equations and solve the system.

Form required: an ordered pair. Altitude from C: x = 2; from A: y = x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle with A(0, 0), B(4, 0), C(0, 3). Where is the orthocenter?

Orthocenter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q2/5

2824

QUESTION ID

S71-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(0, 0)

GRADING NOTE Form required: an ordered pair. Right angle at A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-F1-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q2/5

True or false?

The orthocenter of a right triangle is at the right-angle vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q3/5

2825

QUESTION ID
S71-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Two legs are altitudes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

To find the orthocenter, you need at least

A. three medians

B. the circumcenter

C. one altitude

D. two altitudes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q4/5

2826

QUESTION ID

S71-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. two altitudes

GRADING NOTE Two lines meet in a point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Check that the third altitude of the triangle A(0, 0), B(6, 0), C(2, 4) also passes through (2, 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q5/5

2827

QUESTION ID
S71-F1-Q5

TYPE
Short answer

CORRECT ANSWER
AC has slope 2, so the altitude from B has slope −½: y = −½(x − 6); at x = 2, y = 2. Yes.

GRADING NOTE Any equivalent check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 71 • FORWARD 1 • Q5/5

SET 71 • HSG-GPE.B.5

FORWARD 2

Shortest path to a straight line in a design • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A well is at W(2, 9). A straight canal runs along $y = 2x - 1$ (units: 100 m). Where should a pipe from the well meet the canal so the pipe is as short as possible, and how long is it (to the nearest hundredth of a unit)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q1/5

2829

QUESTION ID

S71-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

At (4.4, 7.8); about 2.68 units (268 m)

GRADING NOTE REQUIRED WORK: Use the perpendicular from W to the canal.

Form required: the point and the length (rounding to 0.01 unit or 1 m). Perpendicular $y = -\frac{1}{2}x + 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A pipe will connect a well to a straight canal. Why is the shortest possible pipe perpendicular to the canal?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q2/5

2830

QUESTION ID
S71-F2-Q2

TYPE
Short answer

CORRECT ANSWER
The perpendicular segment is the shortest distance from a point to a line.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A pipe meeting the canal at a 60° angle is shorter than one meeting it at 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q3/5

2831

QUESTION ID
S71-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The perpendicular is shortest; the slanted pipe is a hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A path from (0, 5) to the road y = 0 is shortest at

A. (0, 0)

B. (5, 0)

C. (−5, 0)

D. (0, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q4/5

2832

QUESTION ID

S71-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. (0, 0)

GRADING NOTE Straight down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A sensor at (3, 4) must connect to the line y = x. Where should the wire meet the line?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q5/5

2833

QUESTION ID

S71-F2-Q5

TYPE

Short answer

CORRECT ANSWER

(3.5, 3.5)

GRADING NOTE Form required: an ordered pair. y = −x + 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 71 • FORWARD 2 • Q5/5

2833

SET 72 • HSG-GPE.B.6 • EXPRESSING GEOMETRIC PROPERTIES WITH EQUATIONS

Partition a directed segment in a given ratio

MAIN	Geometry main question	**HSG-GPE.B.6**
BACKWARD 1	A ratio as a fraction of the whole	**6.RP.A.1**
BACKWARD 2	Horizontal and vertical change	**6.NS.C.8**
BACKWARD 3	Find a fraction of a number	**6.NS.A.1**
FORWARD 1	Work backward: find an endpoint or the ratio	**HSG-GPE.B.6**
FORWARD 2	The centroid as a partition point	**HSG-GPE.B.4**

SET 72 • HSG-GPE.B.6

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

Find the point P on the directed segment from A(1, 2) to B(13, 8) that partitions it in the ratio 1 : 2.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q1/5

2836

QUESTION ID

S72-M-Q1

TYPE

Short answer

CORRECT ANSWER

(5, 4)

GRADING NOTE Form required: an ordered pair. ⅓ of the way from A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the point P on the directed segment from A(−4, 6) to B(6, 1) that partitions it in the ratio 3 : 2.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q2/5

2837

QUESTION ID

S72-M-Q2

TYPE

Short answer

CORRECT ANSWER

(2, 3)

GRADING NOTE Form required: an ordered pair. 3/5 of the way from A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q2/5

Find the point P on the directed segment from A(2, −3) to B(−6, 9) that partitions it in the ratio 1 : 3.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q3/5

2838

QUESTION ID
S72-M-Q3

TYPE
Short answer

CORRECT ANSWER
(0, 0)

GRADING NOTE Form required: an ordered pair. ¼ of the way from A.

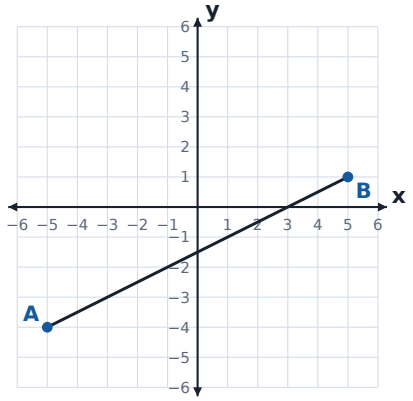
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q3**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q3/5

2838

ANSWER KEY Reduced copy of the question page, with its answer

Find the point P on the directed segment from A to B that partitions it in the ratio 2 : 3.



P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q4/5

2839

QUESTION ID

S72-M-Q4

TYPE

Short answer

CORRECT ANSWER

(−1, −2)

GRADING NOTE Form required: an ordered pair. A(−5, −4), B(5, 1); 2/5 of (10, 5) is (4, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q4/5

Point P partitions the directed segment from B(10, 5) to A(0, 0) in the ratio 1 : 4. P is

A. (8, 4)

B. (2, 1)

C. (5, 2.5)

D. (4, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q5/5

2840

QUESTION ID
S72-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. (8, 4)

GRADING NOTE Start at B: ⅕ of the way toward A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • MAIN • Q5/5

SET 72 • HSG-GPE.B.6

BACKWARD 1

A ratio as a fraction of the whole • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A ratio of 2 : 3 splits a whole into how many equal parts?

Parts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q1/5

2842

QUESTION ID

S72-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a 1 : 2 partition, the first part is what fraction of the whole?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q2/5

2843

QUESTION ID

S72-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{3}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In a 3 : 2 partition, the first part is

A. 3/2 of the whole

B. 2/3 of the whole

C. 3/5 of the whole

D. 2/5 of the whole

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q3/5

2844

QUESTION ID

S72-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3/5 of the whole

GRADING NOTE 3 of 5 parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q3/5

2844

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A ratio of 1 : 3 means the point is ¼ of the way along.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q4/5

2845

GRADING NOTE 1 of 4 parts.

QUESTION ID

S72-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q4/5

True or false?

A ratio of 1 : 2 means the point is halfway.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q5/5

2846

QUESTION ID

S72-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\frac{1}{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 72 • BACKWARD 1 • Q5/5

SET 72 • HSG-GPE.B.6

BACKWARD 2

Horizontal and vertical change • Grade 6 • The Number System • 6.NS.C.8

ANSWER KEY Reduced copy of the question page, with its answer

From (1, 2) to (13, 8), what is the change in x?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q1/5

2848

QUESTION ID

S72-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

From (−4, 6) to (6, 1), what is the change in y?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q2/5

2849

QUESTION ID

S72-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−5

GRADING NOTE Any equivalent form (a decrease of 5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

From (2, −3) to (−6, 9), the change in x is

A. −8

B. −4

C. 4

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q3/5

2850

QUESTION ID

S72-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −8

GRADING NOTE −6 − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q3/5

2850

True or false?

Change in x is the ending x minus the starting x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q4/5

2851

QUESTION ID

S72-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Direction matters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The change in y from (0, 5) to (0, 1) is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q5/5

2852

QUESTION ID

S72-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 72 • BACKWARD 2 • Q5/5

SET 72 • HSG-GPE.B.6

BACKWARD 3

Find a fraction of a number • Grade 6 • The Number System • 6.NS.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Find ⅓ of 12.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q1/5

2854

QUESTION ID

S72-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find 3/5 of −5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q2/5

2855

QUESTION ID

S72-B3-Q2

TYPE

Short answer

CORRECT ANSWER

−3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

2/5 of 10 is

A. 5

B. 25

C. 4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q3/5

2856

QUESTION ID

S72-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4

GRADING NOTE $10 \div 5 \times 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q3/5

2856

True or false?

¼ of −8 is −2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q4/5

2857

QUESTION ID

S72-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

⅔ of 9 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q5/5

2858

QUESTION ID

S72-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 72 • BACKWARD 3 • Q5/5

SET 72 • HSG-GPE.B.6

FORWARD 1

Work backward: find an endpoint or the ratio • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

ANSWER KEY Reduced copy of the question page, with its answer

P(3, 4) partitions the directed segment from A(1, 2) to B in the ratio 1 : 2. Find B.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q1/5

2860

QUESTION ID

S72-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(7, 8)

GRADING NOTE Form required: an ordered pair. AP is ⅓ of AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

P(4, 3) is on the segment from A(0, 1) to B(10, 6). In what ratio does P partition AB?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q2/5

2861

QUESTION ID

S72-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2 : 3

GRADING NOTE Form required: a ratio. 4/10 = 2/5 of the way.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F1-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q2/5

True or false?

The midpoint partitions a segment in the ratio $1 : 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q3/5

2862

QUESTION ID
S72-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Halfway.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q3/5

P partitions AB in 3 : 1. Then P is closer to

A. B

B. neither

C. A

D. the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q4/5

2863

QUESTION ID

S72-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. B

GRADING NOTE ¾ of the way from A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F1-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

M(2, 5) is the midpoint of AB with A(−1, 1). Find B.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q5/5

2864

QUESTION ID

S72-F1-Q5

TYPE

Short answer

CORRECT ANSWER

(5, 9)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F1-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 72 • FORWARD 1 • Q5/5

SET 72 • HSG-GPE.B.6

FORWARD 2

The centroid as a partition point • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Triangle with A(0, 0), B(6, 0), C(3, 9). Find the point ⅔ of the way from A to the midpoint of BC.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q1/5

2866

QUESTION ID

S72-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 3)

GRADING NOTE Form required: an ordered pair. Midpoint (4.5, 4.5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F2-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Show that the centroid (the point 2 : 1 along a median from the vertex) of the triangle with vertices (x_1, y_1), (x_2, y_2), (x_3, y_3) is the average of the vertices.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q2/5

2867

QUESTION ID

S72-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

$((x_1 + x_2 + x_3)/3,$
$(y_1 + y_2 + y_3)/3)$

GRADING NOTE REQUIRED WORK: Partition the median from (x_1, y_1) to the midpoint of the opposite side in the ratio 2 : 1.

Any valid coordinate proof. Essential: midpoint $((x_2 + x_3)/2, (y_2 + y_3)/2)$; $x_1 + \frac{2}{3}((x_2 + x_3)/2 - x_1) = (x_1 + x_2 + x_3)/3$; the same for y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q2/5

True or false?

The centroid of the triangle (0, 0), (6, 0), (3, 9) is (3, 4.5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q3/5

2868

QUESTION ID
S72-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is (3, 3), the average of the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The centroid of (1, 2), (4, 8), (7, −1) is

A. (4, 3)

B. (4, 9)

C. (3, 3)

D. (12, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q4/5

2869

QUESTION ID

S72-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. (4, 3)

GRADING NOTE (12/3, 9/3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the centroid of the triangle with vertices (−2, 5), (4, 1), (7, 0).

Centroid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q5/5

2870

QUESTION ID
S72-F2-Q5

TYPE
Short answer

CORRECT ANSWER
(3, 2)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F2-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 72 • FORWARD 2 • Q5/5

SET 73 • HSG-GPE.B.7 • EXPRESSING GEOMETRIC PROPERTIES WITH EQUATIONS

Compute perimeters of polygons with coordinates

MAIN	Geometry main question	**HSG-GPE.B.7**
BACKWARD 1	Distance between two points	**8.G.B.8**
BACKWARD 2	Lengths of horizontal and vertical sides	**6.G.A.3**
BACKWARD 3	Simplify square roots	**HSN-RN.A.2**
FORWARD 1	Find a missing coordinate from a perimeter	**HSG-GPE.B.7**
FORWARD 2	Fencing and cost from a coordinate plan	**HSG-MG.A.3**

SET 73 • HSG-GPE.B.7

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Find the perimeter of the triangle with vertices (0, 0), (3, 4), and (3, 0).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q1/5

2873

QUESTION ID
S73-M-Q1

TYPE
Short answer

CORRECT ANSWER
12 units

GRADING NOTE Any equivalent form. 5 + 4 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-M-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the perimeter of the quadrilateral with vertices A(1, 1), B(4, 5), C(10, −3), D(7, −7).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q2/5

2874

QUESTION ID
S73-M-Q2

TYPE
Short answer

CORRECT ANSWER
30 units

GRADING NOTE Any equivalent form. 5 + 10 + 5 + 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-M-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q2/5

Find the perimeter of the triangle with vertices (0, 0), (4, 0), (2, 4). Give the exact value and a decimal to the nearest tenth.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q3/5

2875

QUESTION ID

S73-M-Q3

TYPE

Short answer

CORRECT ANSWER

$4 + 4\sqrt{5} \approx 12.9$ units

GRADING NOTE Form required: both. Two sides $\sqrt{20} = 2\sqrt{5}$.

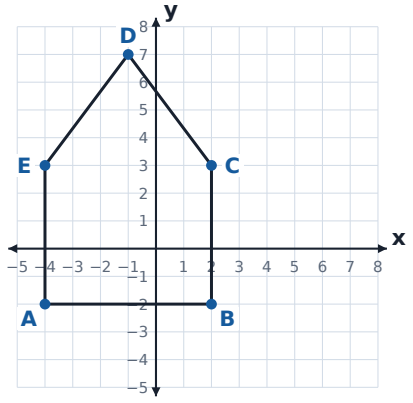
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-M-Q3**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the perimeter of pentagon ABCDE.

Perimeter: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q4/5

2876

QUESTION ID

S73-M-Q4

TYPE

Short answer

CORRECT ANSWER

26 units

GRADING NOTE Any equivalent form. 6 + 5 + 5 + 5 + 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q4/5

A square has vertices (0, 0), (3, 1), (2, 4), (−1, 3). Its perimeter is

A. 4√10

B. 10

C. 40

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q5/5

2877

QUESTION ID
S73-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. 4√10

GRADING NOTE Side √10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • MAIN • Q5/5

SET 73 • HSG-GPE.B.7

BACKWARD 1

Distance between two points • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (2, 3) to (−1, 7)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q1/5

2879

QUESTION ID

S73-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (4, 5) to (10, −3)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q2/5

2880

QUESTION ID

S73-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (0, 0) to (2, 4)?

A. 20

B. 2√5

C. 6

D. √6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q3/5

2881

QUESTION ID

S73-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2√5

GRADING NOTE √20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q3/5

True or false?

The distance from (0, 0) to (3, 1) is √10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q4/5

2882

QUESTION ID

S73-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 9 + 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q4/5

True or false?

Distances can be negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q5/5

2883

QUESTION ID

S73-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Never.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 1 • Q5/5

SET 73 • HSG-GPE.B.7

BACKWARD 2

Lengths of horizontal and vertical sides • Grade 6 • Geometry • 6.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Length of the side from (−4, −2) to (2, −2)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q1/5

2885

QUESTION ID

S73-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Length of the side from (2, −2) to (2, 3)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q2/5

2886

QUESTION ID

S73-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Length from $(-3, 4)$ to $(-3, -1)$?

A. 7

B. 3

C. 5

D. -5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q3/5

2887

QUESTION ID

S73-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5

GRADING NOTE $|4 - (-1)|$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q3/5

True or false?

A horizontal side's length is the absolute value of the difference of the x-coordinates of its endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q4/5

2888

QUESTION ID
S73-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $|x_2 - x_1|$; a length is never negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q4/5

True or false?

The side from (1, 1) to (1, 9) has length 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q5/5

2889

QUESTION ID

S73-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 73 • BACKWARD 2 • Q5/5

SET 73 • HSG-GPE.B.7

BACKWARD 3

Simplify square roots • High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Simplify √20.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q1

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q1/5

2891

QUESTION ID
S73-B3-Q1

TYPE
Short answer

CORRECT ANSWER
2√5

GRADING NOTE Form required: simplest radical form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-B3-Q1**

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify √50.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q2

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q2/5

2892

QUESTION ID
S73-B3-Q2

TYPE
Short answer

CORRECT ANSWER
5√2

GRADING NOTE Form required: simplest radical form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-B3-Q2**

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$2\sqrt{5} + 2\sqrt{5} =$

A. $2\sqrt{10}$

B. $8\sqrt{5}$

C. $4\sqrt{10}$

D. $4\sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q3

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q3/5

2893

QUESTION ID

S73-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $4\sqrt{5}$

GRADING NOTE Like terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-B3-Q3**

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q3/5

True or false?

$\sqrt{10} \approx 3.16$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q4/5

2894

QUESTION ID

S73-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Calculator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q4

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q4/5

True or false?

$\sqrt{9} + \sqrt{16} = \sqrt{25}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q5

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q5/5

2895

QUESTION ID

S73-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3 + 4 = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-B3-Q5**

High School — Algebra I (course placement in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 73 • BACKWARD 3 • Q5/5

SET 73 • HSG-GPE.B.7

FORWARD 1

Find a missing coordinate from a perimeter • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle has vertices (0, 0), (k, 0), (k, 3), (0, 3) with k > 0 and perimeter 22. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q1/5

2897

QUESTION ID

S73-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form. 2k + 6 = 22.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F1-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle with vertices (0, 0), (6, 0), (3, h), h > 0, has perimeter 16. Find h.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q2/5

2898

QUESTION ID

S73-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. Each slanted side is 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F1-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q2/5

True or false?

The triangle with vertices (0, 0), (6, 0), (3, h) has the same perimeter for h = 4 and for h = −4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q3/5

2899

QUESTION ID

S73-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both give slanted sides of 5 (perimeter 16); a condition such as h > 0 is needed to pick one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A square has vertices (0, 0), (a, 0), (a, a), (0, a) with a > 0 and perimeter 18. Find a.

A. 4

B. 3

C. 9

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q4/5

2900

QUESTION ID

S73-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 4.5

GRADING NOTE 4a = 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q4/5

A right triangle has vertices (0, 0), (8, 0), (8, k) with k > 0 and perimeter 24. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q5/5

2901

QUESTION ID

S73-F1-Q5

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. $8 + k + \sqrt{(64 + k^2)} = 24$, so $\sqrt{(64 + k^2)} = 16 - k$; $64 + k^2 = 256 - 32k + k^2$; $k = 6$ (check: $8 + 6 + 10 = 24$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F1-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 73 • FORWARD 1 • Q5/5

SET 73 • HSG-GPE.B.7

FORWARD 2

Fencing and cost from a coordinate plan • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A garden plan has corners (0, 0), (8, 0), (8, 6), (0, 6), in meters. Fencing costs $12 per meter. Find the cost to fence it.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q1/5

2903

QUESTION ID
S73-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$336

GRADING NOTE Any equivalent form. 28 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangular lot has corners (0, 0), (30, 0), (30, 40) in feet. Edging costs $2.50 per foot. Find the total cost.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q2/5

2904

QUESTION ID
S73-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$300

GRADING NOTE Any equivalent form. 120 ft.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q2/5

True or false?

On a map with 1 unit = 5 m, an area of 12 square units is 60 m².

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q3/5

2905

QUESTION ID
S73-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Areas scale by 5²: 12 × 25 = 300 m².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A path has corners (0, 0), (3, 4), (9, 12) (units: 10 m). Its length is

A. 150 m

B. 210 m

C. 15 m

D. 21 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q4/5

2906

QUESTION ID
S73-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 150 m

GRADING NOTE 5 + 10 = 15 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A run around a lot with corners (0, 0), (5, 12), (5, 0) in units of 100 m: how many meters?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q5/5

2907

QUESTION ID
S73-F2-Q5

TYPE
Short answer

CORRECT ANSWER
3000 m

GRADING NOTE Any equivalent form. 13 + 12 + 5 = 30 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 73 • FORWARD 2 • Q5/5

Compute areas of triangles and rectangles with coordinates

MAIN	Geometry main question	**HSG-GPE.B.7**
BACKWARD 1	Area of a triangle from base and height	**6.G.A.1**
BACKWARD 2	Lengths on the coordinate plane	**6.G.A.3**
BACKWARD 3	Distance between two points	**8.G.B.8**
FORWARD 1	Find a missing vertex from an area	**HSG-GPE.B.7**
FORWARD 2	Use area from coordinates with density	**HSG-MG.A.2**

SET 74 • HSG-GPE.B.7

MAIN

Geometry main question • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Find the area of the triangle with vertices (0, 0), (8, 0), and (3, 5).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q1/5

2910

QUESTION ID
S74-M-Q1

TYPE
Short answer

CORRECT ANSWER
20 square units

GRADING NOTE Any equivalent form. Base 8, height 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-M-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q1/5

Rectangle ABCD has A(1, 1), B(5, 3), C(4, 5), D(0, 3).
Find its area.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q2/5

2911

QUESTION ID
S74-M-Q2

TYPE
Short answer

CORRECT ANSWER
10 square units

GRADING NOTE Any equivalent form. AB = √20, BC = √5.

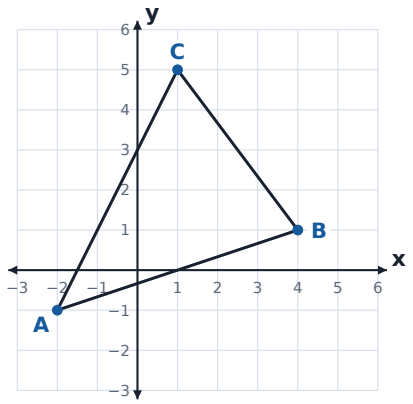
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-M-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the area of triangle ABC.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q3/5

2912

QUESTION ID
S74-M-Q3

TYPE
Short answer

CORRECT ANSWER
15 square units

GRADING NOTE Any equivalent form. Bounding rectangle 36 minus 6, 6, and 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the area of the triangle with vertices A(0, 0), B(6, 2), C(2, 6).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q4/5

2913

QUESTION ID

S74-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

16 square units

GRADING NOTE REQUIRED WORK: Enclose the triangle in a rectangle and subtract the three right triangles.

Any equivalent form. 36 − 6 − 8 − 6. Alternative methods (shoelace) accepted with work.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-M-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q4/5

A triangle has vertices (0, 0), (10, 0), and (x, 4). What is its area?

A. 20, for any x

B. It depends on x

C. 40

D. 20 only if x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q5/5

2914

QUESTION ID

S74-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 20, for any x

GRADING NOTE Base 10, height 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • MAIN • Q5/5

SET 74 • HSG-GPE.B.7

BACKWARD 1

Area of a triangle from base and height • Grade 6 • Geometry • 6.G.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Area of a triangle with base 8 and height 5?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q1/5

2916

GRADING NOTE Any equivalent form.

QUESTION ID

S74-B1-Q1

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Area of a right triangle with legs 6 and 2?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q2/5

2917

QUESTION ID

S74-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A =

A. ½bh

B. b + h

C. 2bh

D. bh

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q3/5

2918

QUESTION ID

S74-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. ½bh

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q3/5

True or false?

The height of a triangle is perpendicular to the base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q4/5

2919

QUESTION ID

S74-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The height must be one of the sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q5/5

2920

QUESTION ID

S74-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only in right triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 74 • BACKWARD 1 • Q5/5

SET 74 • HSG-GPE.B.7

BACKWARD 2

Lengths on the coordinate plane • Grade 6 • Geometry • 6.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Length from (−2, −1) to (4, −1)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q1/5

2922

QUESTION ID

S74-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Height of the point (3, 5) above the x-axis?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q2/5

2923

QUESTION ID

S74-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q2/5

Width of the rectangle enclosing x from −2 to 4?

A. 2

B. 8

C. 4

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q3/5

2924

QUESTION ID

S74-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 6

GRADING NOTE 4 − (−2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q3/5

True or false?

A point with y-coordinate −3 is 3 units from the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q4/5

2925

QUESTION ID
S74-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Absolute value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q4/5

True or false?

The length from (0, 2) to (0, 9) is 11.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q5/5

2926

QUESTION ID

S74-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 74 • BACKWARD 2 • Q5/5

SET 74 • HSG-GPE.B.7

BACKWARD 3

Distance between two points • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (1, 1) to (5, 3)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q1/5

2928

QUESTION ID

S74-B3-Q1

TYPE

Short answer

CORRECT ANSWER

√20

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Distance from (5, 3) to (4, 5)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q2/5

2929

QUESTION ID

S74-B3-Q2

TYPE

Short answer

CORRECT ANSWER

√5

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$\sqrt{20} \cdot \sqrt{5} =$

A. 10

B. 25

C. $\sqrt{25}$

D. 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q3/5

2930

QUESTION ID

S74-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 10

GRADING NOTE $\sqrt{100}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q3/5

2930

True or false?

Slanted rectangle sides need the distance formula.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q4/5

2931

QUESTION ID

S74-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Not grid-aligned.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q4/5

True or false?

$\sqrt{20} \cdot \sqrt{5} = 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q5/5

2932

QUESTION ID

S74-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 74 • BACKWARD 3 • Q5/5

SET 74 • HSG-GPE.B.7

FORWARD 1

Find a missing vertex from an area • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Triangle with vertices (0, 0), (6, 0), (2, k) has area 18.
Find all possible k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q1/5

2934

QUESTION ID
S74-F1-Q1

TYPE
Short answer

CORRECT ANSWER
6 or −6

GRADING NOTE Form required: both values. ½ · 6 · |k| = 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle has vertices (0, 0), (a, 0), (a, 5), (0, 5), $a > 0$, and area 35. Find a.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q2/5

2935

QUESTION ID

S74-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q2/5

True or false?

Moving the top vertex of a triangle parallel to its base does not change its area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q3/5

2936

QUESTION ID

S74-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q3/5

Triangle (0, 0), (4, 0), (x, 6). For which x is the area 12?

A. Any x

B. No x

C. x = 4 only

D. x = 2 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q4/5

2937

QUESTION ID

S74-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Any x

GRADING NOTE ½ · 4 · 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle (1, 1), (7, 1), (4, k) has area 21 and k > 1. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q5/5

2938

QUESTION ID

S74-F1-Q5

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form. Base 6, height 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 74 • FORWARD 1 • Q5/5

SET 74 • HSG-GPE.B.7

FORWARD 2

Use area from coordinates with density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A park on a map has corners (0, 0), (4, 0), (4, 3), (0, 3), with 1 unit = 1 km. 1,800 people live there. Find the population density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q1/5

2940

QUESTION ID

S74-F2-Q1

TYPE

Short answer

CORRECT ANSWER

150 people per km²

GRADING NOTE Any equivalent form. Area 12 km².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangular field has vertices (0, 0), (200, 0), (0, 150) in meters. Seeds are planted at 4 per m². How many seeds?

Seeds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q2/5

2941

QUESTION ID
S74-F2-Q2

TYPE
Short answer

CORRECT ANSWER
60,000

GRADING NOTE Any equivalent form. Area 15,000 m².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Density based on area is an area divided by a quantity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q3/5

2942

QUESTION ID
S74-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is a quantity divided by an area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A lot with corners (0, 0), (6, 0), (6, 5), (0, 5) (meters) holds 60 plants. Density?

A. 2 plants per m²

B. 30 plants per m²

C. 11 plants per m²

D. 0.5 plants per m²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q4/5

2943

QUESTION ID

S74-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2 plants per m²

GRADING NOTE 60/30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle with corners (0, 0), (3, 0), (3, 2), (0, 2) (miles) has 4,200 residents. Find the density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q5/5

2944

QUESTION ID
S74-F2-Q5

TYPE
Short answer

CORRECT ANSWER
700 people per square mile

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 74 • FORWARD 2 • Q5/5

SET 75 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

Informal arguments for the circumference of a circle

MAIN	Geometry main question	**HSG-GMD.A.1**
BACKWARD 1	Circumference formulas	**7.G.B.4**
BACKWARD 4	Perimeters of regular polygons	**3.MD.D.8**
BACKWARD 3	All circles are similar	**HSG-C.A.1**
FORWARD 1	Trap π between inscribed and circumscribed polygons	**HSG-GMD.A.1**
FORWARD 2	Extend the circumference argument to arcs and curved boundari…	**HSG-GMD.A.1**

SET 75 • HSG-GMD.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Regular hexagons are inscribed in and circumscribed about a circle of radius 1. Their perimeters are 6 and $4\sqrt{3} \approx 6.93$. Explain why this shows that $3 < \pi < 2\sqrt{3} \approx 3.46$.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q1/5

2947

QUESTION ID

S75-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

The circle lies between the two hexagons, so $6 < C < 4\sqrt{3}$; with $C = 2\pi$ this gives $3 < \pi < 2\sqrt{3} \approx 3.46$.

GRADING NOTE REQUIRED WORK: Compare the circumference 2π with the two perimeters, then divide by 2.

Any valid explanation. Essential: the inscribed hexagon is six equilateral triangles of side 1 (perimeter 6); the circumference is between the inner and outer perimeters; $C = 2\pi \cdot 1$; divide by 2. Invalid: quoting $\pi \approx 3.14$ as the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the ratio of circumference to diameter is the same number (π) for every circle, so that $C = \pi d = 2\pi r$.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q2/5

2948

QUESTION ID

S75-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Every circle is a dilation of every other, so C and d scale by the same factor and C/d is constant; that constant is called π.

GRADING NOTE

REQUIRED WORK: Use the fact that all circles are similar.

Any valid explanation. Essential: circles are similar (dilation); lengths scale by the same factor k; $C_2/d_2 = kC_1/kd_1 = C_1/d_1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A wheel with diameter 70 cm rolls one full turn along level ground without slipping, and a student measures the distance traveled as 219.9 cm. Explain why that distance is the circumference, use the measurement to estimate π, and say why a measurement supports the formula but does not prove it.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q3/5

2949

QUESTION ID

S75-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

Each point of the rim touches the ground once in one turn, so the distance equals the circumference; 219.9 ÷ 70 ≈ 3.141, close to π. A measurement has error and covers one wheel only.

GRADING NOTE REQUIRED WORK: Relate each point of the rim to the ground it touches; divide distance by diameter.

Any valid explanation. Essential: rolling without slipping lays the rim along the ground; C/d ≈ 3.141; evidence versus proof.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q3/5

2949

ANSWER KEY Reduced copy of the question page, with its answer

A regular hexagon is inscribed in a circle of radius 5. Find the hexagon's perimeter, compare it with the circle's circumference, and explain what happens as the number of sides increases.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q4/5

2950

QUESTION ID

S75-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

Perimeter 30 < 10π ≈ 31.4; with more sides the perimeter gets closer to the circumference.

GRADING NOTE REQUIRED WORK: Split the hexagon into six triangles from the center; compare 30 with 10π; describe the trend.

Any valid explanation. Essential: six equilateral triangles of side 5 (central angles 60°, two radii), so perimeter 30; 30 < 10π; more sides bring the perimeter closer to the circumference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Regular polygons with 6, 12 and 24 sides are inscribed in a circle of radius 1. Their perimeters are 6, 6.212 and 6.265. Explain what these values suggest about the circumference and how the formula C = 2πr follows for a circle of any radius r.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q5/5

2951

QUESTION ID

S75-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

The perimeters increase toward 2π ≈ 6.283, the circumference of the unit circle; a circle of radius r is the unit circle dilated by r, so its circumference is 2π · r.

GRADING NOTE REQUIRED WORK: Describe the trend of the perimeters, then use similarity (scaling by r).

Any valid explanation. Essential: the inscribed perimeters approach the circumference from below; the limit for radius 1 is 2π; dilation by r multiplies lengths by r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • MAIN • Q5/5

SET 75 • HSG-GMD.A.1

BACKWARD 1

Circumference formulas • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the exact circumference of a circle of radius 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q1/5

2953

QUESTION ID

S75-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the exact circumference of a circle of diameter 9.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q2/5

2954

QUESTION ID

S75-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9π

GRADING NOTE Form required: exact value. C = πd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q2/5

Half the circumference of a circle of radius r is

A. πr^2

B. πr

C. $2\pi r$

D. r/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q3/5

2955

QUESTION ID

S75-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. πr

GRADING NOTE ½ · 2πr.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q3/5

True or false?

$C = 2\pi r$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q4/5

2956

QUESTION ID

S75-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q4/5

True or false?

$A = 2\pi r$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q5/5

2957

QUESTION ID

S75-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $A = \pi r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 75 • BACKWARD 1 • Q5/5

SET 75 • HSG-GMD.A.1

BACKWARD 4

Perimeters of regular polygons • Grade 3 • Measurement & Data • 3.MD.D.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the perimeter of a regular hexagon with side 5.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B4-Q1

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q1/5

2959

QUESTION ID

S75-B4-Q1

TYPE

Short answer

CORRECT ANSWER

30

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B4-Q1

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q1/5

2959

ANSWER KEY Reduced copy of the question page, with its answer

Find the perimeter of a regular octagon with side 2.5 cm.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B4-Q2

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q2/5

2960

QUESTION ID

S75-B4-Q2

TYPE

Short answer

CORRECT ANSWER

20 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B4-Q2

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q2/5

2960

ANSWER KEY Reduced copy of the question page, with its answer

A regular 12-gon has perimeter 36. Each side is

A. 432

B. 3

C. 12

D. 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B4-Q3

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q3/5

2961

QUESTION ID

S75-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3

GRADING NOTE 36 ÷ 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B4-Q3

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q3/5

2961

True or false?

A regular polygon with n sides of length s has perimeter ns.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B4-Q4

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q4/5

2962

QUESTION ID

S75-B4-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B4-Q4

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q4/5

True or false?

A regular pentagon with side 4 has perimeter 16.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B4-Q5

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q5/5

2963

QUESTION ID

S75-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $5 \times 4 = 20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B4-Q5

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 75 • BACKWARD 4 • Q5/5

SET 75 • HSG-GMD.A.1

BACKWARD 3

All circles are similar • High School — Geometry • Geometry: Circles • HSG-C.A.1

True or false?

Any circle can be mapped to any other circle by a translation and a dilation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q1/5

2965

QUESTION ID

S75-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Similarity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle of radius 2 is dilated to radius 6. By what factor is the circumference multiplied?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q2/5

2966

QUESTION ID

S75-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q2/5

Circles of radii 3 and 12 have circumferences in the ratio

A. 1 : 16

B. 3 : 4

C. 1 : 4

D. 1 : 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q3/5

2967

QUESTION ID

S75-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1 : 4

GRADING NOTE Scale factor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Why does C/d not depend on the size of the circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q4/5

2968

QUESTION ID

S75-B3-Q4

TYPE

Short answer

CORRECT ANSWER

Both C and d are multiplied by the same scale factor.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A larger circle has a larger value of π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q5/5

2969

QUESTION ID

S75-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE π is constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.1 • Set 75 • BACKWARD 3 • Q5/5

SET 75 • HSG-GMD.A.1

FORWARD 1

Trap π between inscribed and circumscribed polygons • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A circle has radius 1. Find the perimeters of the inscribed square and the circumscribed square, and use them to write bounds for π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q1/5

2971

QUESTION ID

S75-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Inscribed $4\sqrt{2} \approx 5.66$, circumscribed 8; $5.66 < 2\pi < 8$, so $2.83 < \pi < 4$.

GRADING NOTE Form required: both perimeters and the bounds. Inscribed side $\sqrt{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Inscribed and circumscribed regular 12-gons of a circle of radius 1 have perimeters about 6.212 and 6.431. What bounds for π do they give (to the nearest hundredth)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q2/5

2972

QUESTION ID

S75-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$3.11 < \pi < 3.22$

GRADING NOTE Form required: both bounds. Divide each perimeter by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S75-F1-Q2**

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q2/5

True or false?

Using polygons with more sides gives closer bounds for π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q3/5

2973

QUESTION ID

S75-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both perimeters approach the circumference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Why is the circumscribed polygon's perimeter larger than the circumference?

A. It has fewer sides

B. The circle lies inside it, and a curve inside a convex shape is shorter than the shape's boundary

C. Its radius is smaller

D. Its sides are curved

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q4/5

2974

QUESTION ID
S75-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. The circle lies inside it, and a curve inside a convex shape is shorter than the shape's boundary

GRADING NOTE The circle is enclosed by the polygon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Archimedes used 96-gons and found $3\ 10/71 < \pi < 3\ 1/7$. Write these bounds as decimals to the nearest thousandth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q5/5

2975

QUESTION ID

S75-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$3.141 < \pi < 3.143$

GRADING NOTE Form required: both decimals. $3\ 10/71 \approx 3.1408$ and $3\ 1/7 \approx 3.1429$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 1 • Q5/5

SET 75 • HSG-GMD.A.1

FORWARD 2

Extend the circumference argument to arcs and curved boundaries • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Use the circumference formula to explain why an arc with central angle 90° in a circle of radius 6 has length 3π, and find its length to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q1/5

2977

QUESTION ID

S75-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The arc is 90/360 = ¼ of the circle, so its length is ¼ · 2π · 6 = 3π ≈ 9.4.

GRADING NOTE Form required: the fraction-of-the-circle reasoning and the value (rounding to 0.1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the length of an arc is proportional to its central angle, so that an arc of θ° has length (θ/360) · 2πr.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q2/5

2978

QUESTION ID

S75-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Equal central angles cut off congruent arcs (a rotation about the center maps one onto another), so the 360° of the circle split the circumference 2πr in proportion to the angle.

GRADING NOTE Any equivalent explanation using equal angles ↔ equal arcs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q2/5

True or false?

A semicircular region of radius 5 has perimeter 5π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q3/5

2979

QUESTION ID
S75-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The perimeter also includes the diameter: 5π + 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A running track has two straight sides of 100 m and two semicircular ends of radius 30 m. Its length is about

A. 294 m

B. 200 m

C. 388π m

D. 388 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q4/5

2980

QUESTION ID

S75-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 388 m

GRADING NOTE 200 + 60π ≈ 388.5 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

The circumference argument used inscribed polygons whose perimeters approach 2πr. Explain why inscribed broken lines (chords) can be used the same way to measure the length of an arc.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q5/5

2981

QUESTION ID

S75-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Chords that split the arc into smaller and smaller pieces form broken lines whose lengths increase toward the arc length, just as the polygon perimeters approach the circumference.

GRADING NOTE Any equivalent explanation of the limit of chord lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 75 • FORWARD 2 • Q5/5

SET 103 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

Informal arguments for the area of a circle

MAIN	Geometry main question	**HSG-GMD.A.1**
BACKWARD 1	Area of a circle	**7.G.B.4**
BACKWARD 2	Area of a parallelogram	**6.G.A.1**
BACKWARD 3	Scale factors and area	**7.G.A.1**
FORWARD 1	Area of a circle as a limit of regular polygons	**HSG-GMD.A.1**
FORWARD 2	From area to volume: stacking disks	**HSG-GMD.A.1**

SET 103 • HSG-GMD.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A circle of radius r is cut into many equal sectors, which are rearranged as shown. Explain why this shows that the area of the circle is πr^2.

Work space



Not drawn to scale.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q1/5

2984

QUESTION ID

S103-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Area = (half the circumference) × r = $\pi r \cdot r = \pi r^2$.

GRADING NOTE REQUIRED WORK: Describe the base and height of the rearranged shape and what happens as the number of sectors grows.

Any valid explanation. Essential: the shape approaches a parallelogram (rectangle); its height approaches r; its base is half the circumference, πr; area = $\pi r \cdot r = \pi r^2$. Invalid: "the formula says so" or measuring one drawing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle of radius r is cut into many thin rings, like the layers of an onion. Each ring is cut open and straightened into a thin strip, and the strips are stacked from shortest (at the top) to longest (at the bottom). Explain why the stack approaches a triangle and why its area is πr^2.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q2/5

2985

QUESTION ID

S103-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

The strips have lengths $2\pi \cdot$ (distance from the center), from about 0 to $2\pi r$, and the stack is r tall, so it approaches a triangle with base $2\pi r$ and height r: $\frac{1}{2} \cdot 2\pi r \cdot r = \pi r^2$.

GRADING NOTE REQUIRED WORK: Find the lengths of the shortest and longest strips and the height of the stack.

Any valid explanation. Essential: strip length grows in proportion to the distance from the center; outer strip $2\pi r$; height r; triangle area $\frac{1}{2}bh = \pi r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Use similarity to explain why the area of a circle of radius r equals r^2 times the area of a circle of radius 1. Then use the fact that the unit circle has area π to write the formula.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q3/5

2986

QUESTION ID

S103-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

A dilation by r multiplies every area by r^2, so A = r^2 · (area of the unit circle) = πr^2.

GRADING NOTE REQUIRED WORK: A circle of radius r is a dilation of the unit circle by factor r; describe what a dilation does to areas. Any valid explanation. Essential: circles are similar; areas scale by k^2; the unit circle's area is π; A = πr^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A quarter circle of radius 10 is drawn on a grid of unit squares. Counting full squares and estimating the partial ones gives about 78.5 squares. Explain how this supports $A = \pi r^2$ and why counting is evidence rather than proof.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q4/5

2987

QUESTION ID

S103-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

The whole circle covers about $4 \times 78.5 = 314$ squares, and $\pi \cdot 10^2 \approx 314.2$, which agrees. Counting is approximate and checks one circle only, so it supports but does not prove the formula.

GRADING NOTE

REQUIRED WORK: Compare 4×78.5 with πr^2 for $r = 10$.

Any valid explanation. Essential: the scaled-up count (about 314); comparison with 100π; evidence versus proof.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the area of a circle of radius r is more than $2r^2$ and less than $4r^2$, using an inscribed square and a circumscribed square, and check that πr^2 fits these bounds.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q5/5

2988

QUESTION ID

S103-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

The circumscribed square has side 2r (area $4r^2$); the inscribed square has diagonal 2r (area $2r^2$); the circle lies between them, and $2r^2 < \pi r^2 \approx 3.14r^2 < 4r^2$.

GRADING NOTE

REQUIRED WORK: Find the area of each square in terms of r.

Any valid explanation. Essential: both square areas from r (diagonal 2r gives area $\frac{1}{2}(2r)^2$); the circle is between them; $2 < \pi < 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • MAIN • Q5/5

SET 103 • HSG-GMD.A.1

BACKWARD 1

Area of a circle • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the exact area of a circle of radius 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q1/5

2990

QUESTION ID

S103-B1-Q1

TYPE

Short answer

CORRECT ANSWER

16π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the exact area of a circle of diameter 12.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q2/5

2991

QUESTION ID

S103-B1-Q2

TYPE

Short answer

CORRECT ANSWER

36π

GRADING NOTE Form required: exact value. r = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q2/5

2991

ANSWER KEY Reduced copy of the question page, with its answer

A circle has area 25π. Its radius is

A. 5

B. 12.5

C. 25

D. √5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q3/5

2992

QUESTION ID

S103-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

GRADING NOTE $r^2 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A circle of radius 3 has area 9π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q4/5

2993

QUESTION ID

S103-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\pi \cdot 3^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Tripling the radius triples the area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q5/5

2994

QUESTION ID

S103-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The area is multiplied by 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 103 • BACKWARD 1 • Q5/5

SET 103 • HSG-GMD.A.1

BACKWARD 2

Area of a parallelogram • Grade 6 • Geometry • 6.G.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Area of a parallelogram with base 9 and height 4?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q1/5

2996

QUESTION ID

S103-B2-Q1

TYPE

Short answer

CORRECT ANSWER

36

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q1/5

True or false?

A parallelogram has the same area as a rectangle with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q2/5

2997

QUESTION ID

S103-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Cut and slide.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A parallelogram's area is

A. base × slanted side

B. ½ base × height

C. base × height

D. base + height

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q3/5

2998

QUESTION ID

S103-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. base × height

GRADING NOTE Perpendicular height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A parallelogram has base πr and height r. Area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q4/5

2999

QUESTION ID

S103-B2-Q4

TYPE

Short answer

CORRECT ANSWER

πr^2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The slanted side is the height of a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q5/5

3000

QUESTION ID

S103-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Perpendicular height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 103 • BACKWARD 2 • Q5/5

3000

SET 103 • HSG-GMD.A.1

BACKWARD 3

Scale factors and area • Grade 7 • Geometry • 7.G.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A scale drawing uses factor 3. A region of 5 cm² on the original covers how much on the drawing?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B3-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q1/5

3002

QUESTION ID

S103-B3-Q1

TYPE

Short answer

CORRECT ANSWER

45 cm²

GRADING NOTE Any equivalent form. $3^2 \cdot 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B3-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A square of side 2 is enlarged by factor 4. New area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B3-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q2/5

3003

QUESTION ID

S103-B3-Q2

TYPE

Short answer

CORRECT ANSWER

64

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B3-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Enlarging a figure by factor k multiplies its area by

A. k^3

B. k

C. 2k

D. k^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B3-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q3/5

3004

QUESTION ID

S103-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. k^2

GRADING NOTE Both dimensions scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B3-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q3/5

True or false?

Halving every length of a figure multiplies its area by ¼.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B3-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q4/5

3005

QUESTION ID

S103-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(½)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-B3-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q4/5

True or false?

Enlarging by factor 3 triples the area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B3-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q5/5

3006

QUESTION ID

S103-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 9 times.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S103-B3-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 103 • BACKWARD 3 • Q5/5

3006

SET 103 • HSG-GMD.A.1

FORWARD 1

Area of a circle as a limit of regular polygons • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A regular n-gon is inscribed in a circle of radius r. Its area is ½ · perimeter · apothem. Explain why this approaches πr² as n increases.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q1/5

3008

QUESTION ID

S103-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

The perimeter approaches 2πr and the apothem approaches r, so the area approaches ½ · 2πr · r = πr².

GRADING NOTE REQUIRED WORK: Describe what the perimeter and apothem approach.

Any valid explanation. Essential: perimeter → circumference; apothem → radius; substitution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the area of a regular hexagon with side 2 and apothem √3. Compare with the area of its circumscribed circle (radius 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q2/5

3009

QUESTION ID

S103-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$6\sqrt{3} \approx 10.4 < 4\pi \approx 12.6$

GRADING NOTE Form required: both numbers. ½ · 12 · √3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S103-F1-Q2**

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q2/5

True or false?

An inscribed polygon always has less area than its circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q3/5

3010

QUESTION ID

S103-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It lies inside the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q3/5

3010

ANSWER KEY Reduced copy of the question page, with its answer

As n grows, the apothem of an inscribed regular n-gon approaches

A. 0

B. 2r

C. r

D. πr

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q4/5

3011

QUESTION ID

S103-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. r

GRADING NOTE The sides approach the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q4/5

Why do mathematicians call this an "informal" argument?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q5/5

3012

QUESTION ID

S103-F1-Q5

TYPE

Short answer

CORRECT ANSWER

It relies on what the shapes approach (a limit) without a formal proof of the limit.

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 1 • Q5/5

SET 103 • HSG-GMD.A.1

FORWARD 2

From area to volume: stacking disks • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A stack of 40 identical coins, each a disk of radius 1 cm and thickness 0.2 cm, forms a cylinder. Find the exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q1/5

3014

QUESTION ID

S103-F2-Q1

TYPE

Short answer

CORRECT ANSWER

8π cm^3

GRADING NOTE Form required: exact value. $\pi \cdot 1^2 \cdot 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q1/5

True or false?

A cylinder can be thought of as a stack of congruent disks.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q2/5

3015

QUESTION ID

S103-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Layers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q2/5

The volume of a stack of disks of area πr^2 and total height h is

A. $\pi r^2 h$

B. $2\pi rh$

C. πr^2

D. $\frac{1}{3}\pi r^2 h$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q3/5

3016

QUESTION ID

S103-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $\pi r^2 h$

GRADING NOTE Base area × height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how the area of a circle is used to find the volume of a cylinder.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q4/5

3017

QUESTION ID

S103-F2-Q4

TYPE

Short answer

CORRECT ANSWER

The volume is the base area πr^2 times the height h.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Each disk in a stack has area 12 cm² and the stack is 5 cm high. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q5/5

3018

QUESTION ID
S103-F2-Q5

TYPE
Short answer

CORRECT ANSWER
60 cm³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S103-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 103 • FORWARD 2 • Q5/5

SET 76 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

Informal argument for the volume of a cylinder

MAIN	Geometry main question	**HSG-GMD.A.1**
BACKWARD 1	Volume of a rectangular prism as layers	**5.MD.C.5**
BACKWARD 2	Area of a circle	**7.G.B.4**
BACKWARD 3	Cross-sections parallel to the base	**7.G.A.3**
FORWARD 1	Cavalieri's principle with different shapes	**HSG-GMD.A.1**
FORWARD 2	Use the cylinder volume formula to solve problems	**HSG-GMD.A.3**

SET 76 • HSG-GMD.A.1

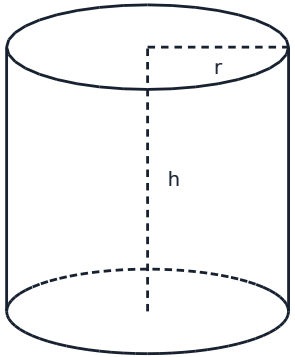
MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the volume of the cylinder shown is $V = \pi r^2 h$.

Work space



Not drawn to scale.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q1/5

3021

QUESTION ID

S76-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Each layer is a disk of area πr^2; stacking layers to height h gives volume (base area) × (height) = $\pi r^2 h$.

GRADING NOTE

REQUIRED WORK: Think of the cylinder as a stack of thin congruent disks.

Any valid explanation. Essential: congruent circular cross-sections of area πr^2; volume = base area × height, as for a prism (layers of unit thickness); result $\pi r^2 h$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A right cylinder and an oblique cylinder have the same radius r and the same height h. Explain why they have the same volume.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q2/5

3022

QUESTION ID

S76-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

At every height both cross-sections are disks of area πr², so by Cavalieri's principle their volumes are equal: πr²h.

GRADING NOTE REQUIRED WORK: Use Cavalieri's principle.

Any valid explanation. Essential: equal heights; equal cross-sectional areas at every level; Cavalieri's principle cited. Invalid: "they look the same size".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder has radius r and height h. A square prism has base area πr^2 and the same height h. Use Cavalieri's principle to explain why the two solids have the same volume, and conclude the cylinder formula.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q3/5

3023

QUESTION ID

S76-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

Every horizontal cross-section of the cylinder is a disk of area πr^2, and every cross-section of the prism has area πr^2; equal heights and equal cross-section areas give equal volumes, so $V = (\pi r^2)h$.

GRADING NOTE REQUIRED WORK: Compare the cross-section areas at every height.

Any valid explanation. Essential: both cross-section areas named and equal at every level; equal heights; Cavalieri's principle; prism volume Bh.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why V = Bh works for both prisms and cylinders.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q4/5

3024

QUESTION ID

S76-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

Both have congruent cross-sections of area B at every height, so the volume is B times the height.

GRADING NOTE

REQUIRED WORK: Describe the cross-sections of each solid and how the volume is built from them.

Any valid explanation. Essential: congruent cross-sections parallel to the base; layers of thickness 1 (or thin layers) each of volume B; h layers give Bh.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder of radius r and height h is approximated by prisms whose bases are regular polygons inscribed in its base circle. Explain why the prism volumes approach $\pi r^2 h$.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q5/5

3025

QUESTION ID

S76-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

Each prism has volume (polygon area) × h; as the number of sides grows, the polygon area approaches πr^2, so the volumes approach $\pi r^2 h$.

GRADING NOTE REQUIRED WORK: Describe what the polygon base areas approach as the number of sides grows.

Any valid explanation. Essential: prism volume Bh; inscribed polygon areas approach the circle area; conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • MAIN • Q5/5

SET 76 • HSG-GMD.A.1

BACKWARD 1

Volume of a rectangular prism as layers • Grade 5 • Measurement & Data • 5.MD.C.5

ANSWER KEY Reduced copy of the question page, with its answer

A prism has a 4 × 3 base and height 5. How many unit cubes are in one layer, and what is the volume?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q1/5

3027

QUESTION ID

S76-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12 per layer; 60

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q1/5

True or false?

V = Bh, where B is the area of the base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q2/5

3028

QUESTION ID

S76-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Layers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A prism has base area 20 and height 7. Volume?

A. 27

B. 140

C. 200

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q3/5

3029

QUESTION ID

S76-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 140

GRADING NOTE B × h.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q3/5

3029

ANSWER KEY Reduced copy of the question page, with its answer

A prism has 6 layers of 10 cubes each. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q4/5

3030

QUESTION ID

S76-B1-Q4

TYPE

Short answer

CORRECT ANSWER

60 cubic units

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q4/5

3030

True or false?

Volume is measured in square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q5/5

3031

QUESTION ID

S76-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Cubic units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 76 • BACKWARD 1 • Q5/5

SET 76 • HSG-GMD.A.1

BACKWARD 2

Area of a circle • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of a circle of radius 3?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q1/5

3033

QUESTION ID

S76-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q1/5

3033

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of a circle of diameter 10?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q2/5

3034

QUESTION ID

S76-B2-Q2

TYPE

Short answer

CORRECT ANSWER

25π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A =

A. 2πr

B. πr

C. πr^2

D. πd

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q3/5

3035

QUESTION ID

S76-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. πr^2

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q3/5

True or false?

The base of a cylinder is a circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q4/5

3036

QUESTION ID

S76-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q4/5

True or false?

The area of a circle of radius 2 is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q5/5

3037

QUESTION ID

S76-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 4π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 76 • BACKWARD 2 • Q5/5

SET 76 • HSG-GMD.A.1

BACKWARD 3

Cross-sections parallel to the base • Grade 7 • Geometry • 7.G.A.3

True or false?

A slice of a right prism parallel to its base is congruent to the base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q1/5

3039

QUESTION ID

S76-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same shape and size.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q1/5

A cylinder is sliced parallel to its base. Describe the cross-section.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q2/5

3040

QUESTION ID

S76-B3-Q2

TYPE

Short answer

CORRECT ANSWER

A circle congruent to the base

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q2/5

A rectangular prism sliced parallel to its 3 × 4 base gives a

A. smaller rectangle

B. 3 × 4 rectangle

C. triangle

D. circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q3/5

3041

QUESTION ID

S76-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3 × 4 rectangle

GRADING NOTE Congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q3/5

Why do all horizontal slices of a cylinder have the same area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q4/5

3042

QUESTION ID

S76-B3-Q4

TYPE

Short answer

CORRECT ANSWER

They are all congruent to the base.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q4/5

True or false?

Horizontal slices of a cylinder get smaller toward the top.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q5/5

3043

QUESTION ID

S76-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE That is a cone.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 76 • BACKWARD 3 • Q5/5

SET 76 • HSG-GMD.A.1

FORWARD 1

Cavalieri's principle with different shapes • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A cylinder with base area 50 cm² and a square prism with base area 50 cm² both have height 8 cm. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q1/5

3045

QUESTION ID
S76-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Equal: 400 cm³ each

GRADING NOTE Form required: the comparison with the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

An oblique cylinder has radius 3 and vertical height 7 (its slanted side is 9). Find its exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q2/5

3046

QUESTION ID

S76-F1-Q2

TYPE

Short answer

CORRECT ANSWER

63π

GRADING NOTE Form required: exact value. Use the vertical height, not the slant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q2/5

True or false?

Cavalieri's principle requires equal cross-sectional areas at every height and equal heights.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q3/5

3047

QUESTION ID

S76-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Statement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q3/5

Two stacks of 30 identical cards, one straight and one slanted, have

A. different volumes

B. equal volumes

C. equal surface areas

D. no volume

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q4/5

3048

QUESTION ID

S76-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. equal volumes

GRADING NOTE Same cross-sections.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q4/5

3048

ANSWER KEY Reduced copy of the question page, with its answer

State Cavalieri's principle in your own words.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q5/5

3049

QUESTION ID
S76-F1-Q5

TYPE
Short answer

CORRECT ANSWER
Two solids with the same height and equal cross-sectional areas at every level have the same volume.

GRADING NOTE Any equivalent statement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 76 • FORWARD 1 • Q5/5

SET 76 • HSG-GMD.A.1

FORWARD 2

Use the cylinder volume formula to solve problems • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A can has radius 4 cm and height 12 cm. Find its volume to the nearest cubic centimeter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q1/5

3051

QUESTION ID
S76-F2-Q1

TYPE
Short answer

CORRECT ANSWER
603 cm³

GRADING NOTE Rounding to the nearest whole number required. 192π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A pipe 2 m long has inner radius 0.1 m. Find the volume of water it holds, to the nearest thousandth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q2/5

3052

QUESTION ID
S76-F2-Q2

TYPE
Short answer

CORRECT ANSWER
0.063 m³

GRADING NOTE Rounding to 0.001 required. 0.02π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q2/5

3052

True or false?

Doubling the height of a cylinder multiplies its volume by 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q3/5

3053

QUESTION ID

S76-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Volume is linear in h, so it doubles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Doubling the radius of a cylinder multiplies its volume by

A. 4

B. 8

C. 6

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q4/5

3054

GRADING NOTE r^2.

QUESTION ID

S76-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylindrical tank has diameter 6 ft and height 10 ft. Exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q5/5

3055

QUESTION ID
S76-F2-Q5

TYPE
Short answer

CORRECT ANSWER
90π ft^3

GRADING NOTE Form required: exact value. r = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 76 • FORWARD 2 • Q5/5

SET 77 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

Informal arguments for the volume of a pyramid

MAIN	Geometry main question	**HSG-GMD.A.1**
BACKWARD 1	Volume of a prism	**7.G.B.6**
BACKWARD 2	Base area of a pyramid	**6.G.A.1**
BACKWARD 4	Split a volume into equal parts	**5.MD.C.5**
FORWARD 1	Cavalieri's principle for pyramids	**HSG-GMD.A.1**
FORWARD 2	Use the pyramid volume formula	**HSG-GMD.A.3**

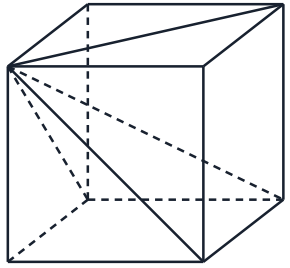
SET 77 • HSG-GMD.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A cube with edge s is split into three congruent pyramids that share one vertex of the cube as their apex, as shown. Use this, together with scaling and Cavalieri's principle, to explain why EVERY pyramid with base area B and height h has volume ⅓Bh.



Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q1/5

3058

QUESTION ID

S77-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Each of the three pyramids has volume ⅓s³ = ⅓(s²)(s). Stretching vertically by h/s multiplies the volume by h/s, giving ⅓s²h. Any pyramid with base area B and height h has cross-sections B((h − y)/h)² at height y, the same as a square pyramid with base area B and height h, so by Cavalieri's principle its volume is ⅓Bh.

GRADING NOTE REQUIRED WORK: (1) Find the volume of one of the three pyramids. (2) Stretch it vertically to any height h. (3) Use Cavalieri's principle to compare any pyramid with a square pyramid of the same base area and height.

Any valid explanation. Essential: (1) the three congruent pyramids fill the cube, each with base s² and height s, so ⅓s³; (2) a vertical stretch extends this to any height; (3) cross-sections are scaled copies of the base with area B((h − y)/h)², so Cavalieri's principle extends it to any base shape and apex position. The cube case alone is NOT sufficient for full credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangular prism with base area B and height h is cut into three pyramids (tetrahedra), as Euclid did. Pairs of these pyramids have congruent bases and equal heights. Explain why each pyramid has volume ⅓Bh.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q2/5

3059

QUESTION ID

S77-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Pyramid 1 and pyramid 2 have equal bases and heights, and so do pyramids 2 and 3, so all three have equal volume. They fill the prism (volume Bh), so each has ⅓Bh; pyramid 1 has the prism's base and height.

GRADING NOTE REQUIRED WORK: Use the fact that pyramids with equal base areas and equal heights have equal volumes (Cavalieri's principle), applied to two pairs.

Any valid explanation. Essential: two equal-volume pairs give three equal volumes; their sum is Bh; one of them has base B and height h, so V = ⅓Bh.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In a class experiment, a hollow square pyramid and a hollow square prism have the same base and the same height. It takes three pyramids full of sand to fill the prism. Explain what this shows about the pyramid formula and why it is evidence, not a proof.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q3/5

3060

QUESTION ID

S77-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

The pyramid holds about ⅓ of the prism, Bh, which supports V = ⅓Bh. The experiment has measurement error and tests one pyramid only, so it does not prove the formula for all pyramids.

GRADING NOTE REQUIRED WORK: Relate the prism volume Bh to the three pyramid fillings, then discuss measurement.

Any valid explanation. Essential: the ⅓ relationship with the prism; evidence versus proof.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cube with edge 6 is split into three congruent pyramids that share one vertex of the cube as their apex. Find the volume of each pyramid, then show that ⅓Bh gives the same value for one pyramid, and explain why this agreement is a check, not a proof for all pyramids.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q4/5

3061

QUESTION ID

S77-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

Each is 216 ÷ 3 = 72; one pyramid has base 6 × 6 = 36 and height 6, and ⅓ · 36 · 6 = 72. One numerical case cannot cover every pyramid.

GRADING NOTE

REQUIRED WORK: Divide the cube's volume by 3; then compute ⅓Bh for one pyramid.

Any valid explanation. Essential: 216 ÷ 3 = 72; B = 36 and h = 6 for one pyramid; ⅓Bh = 72; a single case is evidence only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A square pyramid has base side s and height h. Imagine it built from n stacked square slabs, each of thickness h/n, where the slab k steps from the top has side ks/n. The total slab volume is $(s^2h/n^3) \times$ (the sum of the squares 1^2, 2^2, 3^2, and so on up to n^2). For n = 10 and n = 100 this equals $0.385s^2h$ and $0.338s^2h$. Explain how this supports $V = \frac{1}{3}s^2h$.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q5/5

3062

QUESTION ID

S77-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

The slabs approach the pyramid as they get thinner, and the coefficients 0.385, 0.338, … approach ⅓, so the volume is $\frac{1}{3}s^2h$.

GRADING NOTE

REQUIRED WORK: Describe what the slabs approach and what the coefficients approach as n grows.

Any valid explanation. Essential: the slabs approximate the pyramid more closely as n grows; the coefficient trend toward ⅓; conclusion (an informal limit argument).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • MAIN • Q5/5

SET 77 • HSG-GMD.A.1

BACKWARD 1

Volume of a prism • Grade 7 • Geometry • 7.G.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A prism has base area 36 and height 6. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q1/5

3064

QUESTION ID

S77-B1-Q1

TYPE

Short answer

CORRECT ANSWER

216

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cube has side 6. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q2/5

3065

QUESTION ID

S77-B1-Q2

TYPE

Short answer

CORRECT ANSWER

216

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

V =

A. Bh

B. ⅓Bh

C. 2Bh

D. B + h

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q3/5

3066

QUESTION ID

S77-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Bh

GRADING NOTE Prism.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q3/5

True or false?

A cube is a prism.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q4/5

3067

QUESTION ID

S77-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Square base.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q4/5

True or false?

A prism's volume depends on the number of its faces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q5/5

3068

QUESTION ID
S77-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Base area and height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 77 • BACKWARD 1 • Q5/5

SET 77 • HSG-GMD.A.1

BACKWARD 2

Base area of a pyramid • Grade 6 • Geometry • 6.G.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A square base has side 10. Area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q1/5

3070

QUESTION ID

S77-B2-Q1

TYPE

Short answer

CORRECT ANSWER

100

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangular base has base 6 and height 4. Area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q2/5

3071

QUESTION ID

S77-B2-Q2

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q2/5

The base of a square pyramid is a

A. triangle

B. square

C. rectangle that is not a square

D. circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q3/5

3072

QUESTION ID

S77-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. square

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q3/5

True or false?

B in V = ⅓Bh means the area of the base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q4/5

3073

GRADING NOTE Area.

QUESTION ID

S77-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q4/5

True or false?

B is the length of a base edge.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q5/5

3074

QUESTION ID

S77-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 77 • BACKWARD 2 • Q5/5

SET 77 • HSG-GMD.A.1

BACKWARD 4

Split a volume into equal parts • Grade 5 • Measurement & Data • 5.MD.C.5

ANSWER KEY Reduced copy of the question page, with its answer

A box of volume 60 cm³ is split into three equal parts. Volume of each?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B4-Q1

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q1/5

3076

QUESTION ID

S77-B4-Q1

TYPE

Short answer

CORRECT ANSWER

20 cm³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B4-Q1

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Three congruent blocks fill a cube of volume 27. Volume of one block?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B4-Q2

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q2/5

3077

QUESTION ID

S77-B4-Q2

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B4-Q2

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A solid made of non-overlapping parts has volume

A. the largest part's volume

B. the sum of the parts' volumes

C. the product of the parts' volumes

D. the average volume

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B4-Q3

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q3/5

3078

QUESTION ID

S77-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. the sum of the parts' volumes

GRADING NOTE Additivity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B4-Q3

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q3/5

True or false?

If three equal pieces fill a prism, each piece has ⅓ of the prism's volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B4-Q4

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q4/5

3079

QUESTION ID

S77-B4-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal shares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B4-Q4

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Cutting a solid into pieces changes its total volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B4-Q5

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q5/5

3080

QUESTION ID

S77-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Volume is additive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B4-Q5

Grade 5 • Measurement & Data • 5.MD.C.5 • Set 77 • BACKWARD 4 • Q5/5

3080

SET 77 • HSG-GMD.A.1

FORWARD 1

Cavalieri's principle for pyramids • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A square pyramid and a triangular pyramid both have base area 50 cm² and height 9 cm. Compare their volumes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q1/5

3082

QUESTION ID

S77-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Equal: 150 cm³ each

GRADING NOTE Form required: the comparison with the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

An oblique pyramid has a square base with side 4 and vertical height 6. Find its volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q2/5

3083

QUESTION ID
S77-F1-Q2

TYPE
Short answer

CORRECT ANSWER
32

GRADING NOTE Any equivalent form. ⅓ · 16 · 6; use the vertical height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q2/5

3083

True or false?

Two pyramids with equal base areas and equal heights have equal volumes even if their apexes are in different positions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q3/5

3084

QUESTION ID

S77-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Cavalieri.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q3/5

Why do two pyramids with equal base areas and equal heights have equal cross-section areas at every level?

A. Their bases have the same shape

B. Both cross-sections are scaled copies of the bases by the same factor

C. They have the same slant height

D. They have the same apex

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q4/5

3085

QUESTION ID
S77-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. Both cross-sections are scaled copies of the bases by the same factor

GRADING NOTE Each cross-section at height y is the base scaled by (h − y)/h.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A cross-section of a pyramid halfway up has what fraction of the base area?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q5/5

3086

QUESTION ID
S77-F1-Q5

TYPE
Short answer

CORRECT ANSWER
¼

GRADING NOTE Any equivalent form. Scale factor ½.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 77 • FORWARD 1 • Q5/5

SET 77 • HSG-GMD.A.1

FORWARD 2

Use the pyramid volume formula • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A square pyramid has base side 9 m and height 10 m. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q1/5

3088

QUESTION ID
S77-F2-Q1

TYPE
Short answer

CORRECT ANSWER
270 m³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q1/5

3088

ANSWER KEY Reduced copy of the question page, with its answer

A rectangular pyramid has a 6 in. by 4 in. base and height 7.5 in. Find its volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q2/5

3089

QUESTION ID
S77-F2-Q2

TYPE
Short answer

CORRECT ANSWER
60 in³

GRADING NOTE Any equivalent form. ⅓ · 24 · 7.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q2/5

3089

True or false?

A pyramid with the same base and height as a prism of volume 90 has volume 45.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q3/5

3090

QUESTION ID

S77-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has ⅓ of the volume: 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q3/5

3090

ANSWER KEY Reduced copy of the question page, with its answer

A pyramid with base area 48 and height 5 has volume

A. 120

B. 53

C. 240

D. 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q4/5

3091

QUESTION ID

S77-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 80

GRADING NOTE $\frac{1}{3} \cdot 240$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q4/5

3091

ANSWER KEY Reduced copy of the question page, with its answer

A paperweight is a pyramid whose triangular base has base 6 cm and height 5 cm; the pyramid is 8 cm tall. Find its volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q5/5

3092

QUESTION ID
S77-F2-Q5

TYPE
Short answer

CORRECT ANSWER
40 cm³

GRADING NOTE Any equivalent form. B = 15; ⅓ · 15 · 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 77 • FORWARD 2 • Q5/5

SET 104 • HSG-GMD.A.1 • GEOMETRIC MEASUREMENT AND DIMENSION

Informal arguments for the volume of a cone

MAIN	Geometry main question	**HSG-GMD.A.1**
BACKWARD 1	Area of a circle	**7.G.B.4**
BACKWARD 2	The cylinder volume argument	**HSG-GMD.A.1**
BACKWARD 3	The pyramid volume formula	**HSG-GMD.A.1**
FORWARD 1	Volumes of frustums (cones with the top cut off)	**HSG-GMD.A.3**
FORWARD 2	Use the cone volume formula	**HSG-GMD.A.3**

SET 104 • HSG-GMD.A.1

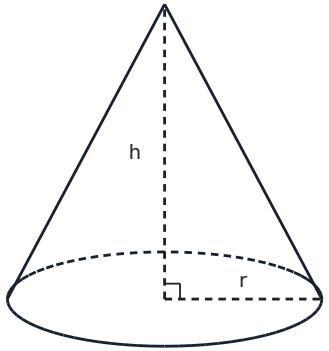
MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the volume of a cone is ⅓πr²h, using pyramids.

Work space



Not drawn to scale.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q1/5

3095

QUESTION ID

S104-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Pyramids with regular n-gon bases inscribed in the circle have V = ⅓Bh; as n grows, B approaches πr² and the pyramids approach the cone, so V = ⅓πr²h.

GRADING NOTE REQUIRED WORK: Think of pyramids whose bases are regular polygons with more and more sides.

Any valid explanation. Essential: pyramid volume ⅓Bh; base polygons approach the circle (B → πr²); the pyramids approach the cone. Alternative: Cavalieri comparison with a pyramid of equal base area and height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone has base radius r and height h. A square pyramid has base area πr² and the same height h. Use Cavalieri's principle to explain why the cone's volume is ⅓πr²h.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q2/5

3096

QUESTION ID

S104-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

At height y both cross-sections are copies of the bases scaled by (h − y)/h, so both have area πr²((h − y)/h)²; equal areas at every level and equal heights give equal volumes, ⅓πr²h.

GRADING NOTE REQUIRED WORK: Compare the cross-section areas of the two solids at the same height y.

Any valid explanation. Essential: the scale factor (h − y)/h for both solids; equal cross-section areas; Cavalieri's principle; the pyramid formula ⅓Bh with B = πr².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In an experiment, three cones full of water exactly fill a cylinder with the same base radius and height. Explain what this shows about the cone formula and why it is evidence, not a proof.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q3/5

3097

QUESTION ID

S104-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

One cone holds about ⅓ of πr²h, which supports V = ⅓πr²h; the experiment has measurement error and tests one cone, so it does not prove the formula.

GRADING NOTE REQUIRED WORK: Relate the cylinder volume πr²h to the three cone fillings, then discuss measurement.

Any valid explanation. Essential: the ⅓ relationship with the cylinder; evidence versus proof.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone with radius r and height h is approximated by n stacked disks of thickness h/n; the disk k steps from the top has radius kr/n. The total disk volume is $(\pi r^2 h/n^3) \times$ (the sum of the squares 1^2, 2^2, 3^2, and so on up to n^2), which equals $0.385\pi r^2 h$ for n = 10 and $0.338\pi r^2 h$ for n = 100. Explain how this supports $V = \frac{1}{3}\pi r^2 h$.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q4/5

3098

QUESTION ID

S104-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

As the disks get thinner they fit the cone more closely, and the coefficients approach ⅓, so $V = \frac{1}{3}\pi r^2 h$.

GRADING NOTE REQUIRED WORK: Describe what the disks approach and what the coefficients approach.

Any valid explanation. Essential: the disks approximate the cone better as n grows; the trend toward ⅓; conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone has base radius r and height h. A plane parallel to the base cuts the cone at height y above the base. Explain why the cross-section is a circle of radius r(h − y)/h, and why this fact is what makes the cone's volume ⅓ of the cylinder's rather than ½.

Work space

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q5/5

3099

QUESTION ID

S104-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

The small cone above the cut is similar to the whole cone with ratio (h − y)/h, so the radius is r(h − y)/h. The areas shrink like the SQUARE of that ratio, $\pi r^2((h - y)/h)^2$, so most of the volume is near the base and the total is $\frac{1}{3}\pi r^2 h$, less than the ½ a linear shrink would suggest.

GRADING NOTE REQUIRED WORK: Use similar triangles in a vertical cross-section through the apex; compare how the cross-section areas shrink.

Any valid explanation. Essential: similar triangles give the radius; area scales by the square of the ratio; link to the ⅓ factor (via Cavalieri or slab sums).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • MAIN • Q5/5

SET 104 • HSG-GMD.A.1

BACKWARD 1

Area of a circle • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of a circle of radius 7?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q1/5

3101

QUESTION ID

S104-B1-Q1

TYPE

Short answer

CORRECT ANSWER

49π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of a circle of diameter 8?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q2/5

3102

QUESTION ID

S104-B1-Q2

TYPE

Short answer

CORRECT ANSWER

16π

GRADING NOTE Form required: exact value. r = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The base of a cone is a

A. triangle

B. sector

C. square

D. circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q3/5

3103

QUESTION ID

S104-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. circle

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q3/5

True or false?

The base area of a cone with radius r is πr^2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q4/5

3104

QUESTION ID

S104-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Circle area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S104-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q4/5

True or false?

The base area of a cone with radius 2 is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q5/5

3105

QUESTION ID

S104-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 4π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 104 • BACKWARD 1 • Q5/5

SET 104 • HSG-GMD.A.1

BACKWARD 2

The cylinder volume argument • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

True or false?

A cylinder's volume is $\pi r^2 h$ because it is a stack of disks of area πr^2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q1/5

3107

QUESTION ID

S104-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Set 76.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Cylinder radius 3, height 5: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q2/5

3108

QUESTION ID

S104-B2-Q2

TYPE

Short answer

CORRECT ANSWER

45π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone fits inside a cylinder with the same base and height. The cone's volume is

A. ½ of the cylinder's

B. ⅓ of the cylinder's

C. ⅔ of the cylinder's

D. equal to the cylinder's

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q3/5

3109

QUESTION ID

S104-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. ⅓ of the cylinder's

GRADING NOTE Cone vs cylinder.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder holds 60 cm³. A cone with the same base and height holds

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q4/5

3110

QUESTION ID

S104-B2-Q4

TYPE

Short answer

CORRECT ANSWER

20 cm³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q4/5

3110

True or false?

A cone is a cylinder with the top cut off flat.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q5/5

3111

QUESTION ID

S104-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It comes to a point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 2 • Q5/5

SET 104 • HSG-GMD.A.1

BACKWARD 3

The pyramid volume formula • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A pyramid has base area 30 and height 8. Volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B3-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q1/5

3113

QUESTION ID

S104-B3-Q1

TYPE

Short answer

CORRECT ANSWER

80

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B3-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q1/5

True or false?

V = ⅓Bh holds for every pyramid, whatever the shape of its base.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B3-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q2/5

3114

QUESTION ID

S104-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Set 77.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B3-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q2/5

A pyramid and a prism have the same base and height. The pyramid's volume is

A. ⅓ of the prism's

B. ½ of the prism's

C. 3 times the prism's

D. equal to the prism's

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B3-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q3/5

3115

QUESTION ID

S104-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. ⅓ of the prism's

GRADING NOTE ⅓Bh.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B3-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A pyramid has volume 40 and height 6. Base area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B3-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q4/5

3116

QUESTION ID

S104-B3-Q4

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE Any equivalent form. 3 · 40 ÷ 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B3-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q4/5

True or false?

A pyramid has half the volume of the prism with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B3-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q5/5

3117

QUESTION ID

S104-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE One third.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-B3-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 104 • BACKWARD 3 • Q5/5

SET 104 • HSG-GMD.A.1

FORWARD 1

Volumes of frustums (cones with the top cut off) • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A cone has radius 6 and height 12. The top cone of height 4 is cut off by a plane parallel to the base. Find the exact volume of the remaining frustum.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q1/5

3119

QUESTION ID

S104-F1-Q1

TYPE

Short answer

CORRECT ANSWER

416π/3

GRADING NOTE Form required: exact value. Whole cone 144π; small cone radius 2, volume 16π/3; 144π − 16π/3 = 416π/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q1/5

3119

ANSWER KEY Reduced copy of the question page, with its answer

A cup is a frustum with bottom radius 3 cm, top radius 4 cm and height 6 cm. Extend it to a full cone and find the cup's volume to the nearest cubic centimeter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q2/5

3120

QUESTION ID

S104-F1-Q2

TYPE

Short answer

CORRECT ANSWER

232 cm³

GRADING NOTE Rounding to the nearest whole number required. Full cone height 24 (radius 4); removed cone height 18 (radius 3); ⅓π(16 · 24 − 9 · 18) = 74π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q2/5

3120

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A frustum's volume is the large cone's volume minus the small cone's volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q3/5

3121

QUESTION ID
S104-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Subtraction of non-overlapping parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone is cut halfway up its height, parallel to the base. The small top cone has what fraction of the whole cone's volume?

A. ⅓

B. ½

C. ⅛

D. ¼

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q4/5

3122

QUESTION ID

S104-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. ⅛

GRADING NOTE $(½)^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why cutting a cone halfway up leaves 7/8 of its volume in the bottom part.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q5/5

3123

QUESTION ID
S104-F1-Q5

TYPE
Short answer

CORRECT ANSWER
The top cone is similar with ratio ½, so its volume is (½)³ = ⅛ of the whole; 1 − ⅛ = 7/8 remains.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 1 • Q5/5

SET 104 • HSG-GMD.A.1

FORWARD 2

Use the cone volume formula • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A cone has radius 3 in. and height 8 in. Find its volume to the nearest tenth.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q1/5

3125

QUESTION ID

S104-F2-Q1

TYPE

Short answer

CORRECT ANSWER

75.4 in³

GRADING NOTE Rounding to 0.1 required. 24π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S104-F2-Q1**

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone-shaped cup has diameter 6 cm and height 9 cm. Exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q2/5

3126

QUESTION ID

S104-F2-Q2

TYPE

Short answer

CORRECT ANSWER

27π cm^3

GRADING NOTE Form required: exact value. r = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q2/5

True or false?

A cone with the same base and height as a cylinder of volume 90 has volume 45.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q3/5

3127

QUESTION ID

S104-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has ⅓ of the volume: 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone has base area 48 and height 5. Its volume is

A. 240

B. 53

C. 80

D. 120

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q4/5

3128

QUESTION ID

S104-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 80

GRADING NOTE $\frac{1}{3} \cdot 240$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A pile of sand is a cone 2 m tall with base radius 3 m. Find its volume to the nearest tenth of a cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q5/5

3129

QUESTION ID

S104-F2-Q5

TYPE

Short answer

CORRECT ANSWER

18.8 m^3

GRADING NOTE Rounding to 0.1 required. 6π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S104-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 104 • FORWARD 2 • Q5/5

SET 78 • HSG-GMD.A.3 • GEOMETRIC MEASUREMENT AND DIMENSION

Use volume formulas for cylinders, pyramids, and cones

MAIN	Geometry main question	**HSG-GMD.A.3**
BACKWARD 1	Area of a circle	**7.G.B.4**
BACKWARD 2	Find a missing side with the Pythagorean Theorem	**8.G.B.7**
BACKWARD 3	Evaluate a formula and round	**7.EE.B.3**
BACKWARD 4	Which solid uses ⅓?	**HSG-GMD.A.1**
FORWARD 1	Composite and hollow solids	**HSG-GMD.A.3**
FORWARD 2	Mass from volume and density	**HSG-MG.A.2**

SET 78 • HSG-GMD.A.3

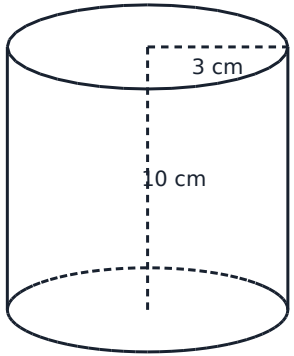
MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder has radius 3 cm and height 10 cm. Find its volume, exact and to the nearest tenth.

Volume: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q1/5

3132

QUESTION ID

S78-M-Q1

TYPE

Short answer

CORRECT ANSWER

90π cm³ ≈ 282.7 cm³

GRADING NOTE Form required: both the exact value and rounding to 0.1, with units.

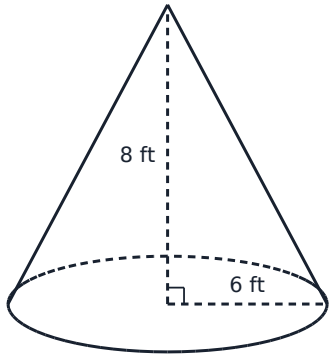
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone has radius 6 ft and height 8 ft. Find its volume, exact and to the nearest tenth.

Volume: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q2/5

3133

QUESTION ID

S78-M-Q2

TYPE

Short answer

CORRECT ANSWER

96π ft³ ≈ 301.6 ft³

GRADING NOTE Form required: both, with units. ⅓π · 36 · 8.

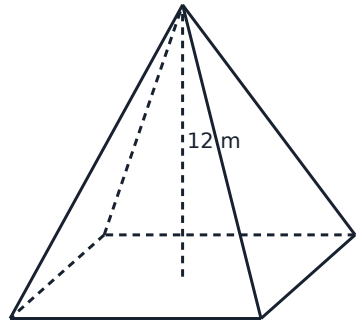
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A square pyramid has base side 10 m and height 12 m. Find its volume.

Volume: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q3/5

3134

QUESTION ID

S78-M-Q3

TYPE

Short answer

CORRECT ANSWER

400 m³

GRADING NOTE Any equivalent form, with units. ⅓ · 100 · 12.

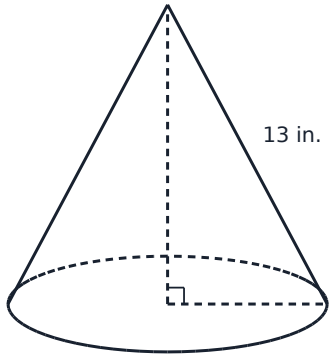
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone has diameter 10 in. and slant height 13 in. Find its volume, exact and to the nearest tenth.

Volume: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q4/5

3135

QUESTION ID

S78-M-Q4

TYPE

Short answer

CORRECT ANSWER

100π in^3 $\approx$ 314.2 in^3

GRADING NOTE Form required: both, with units. r = 5; height √(169 − 25) = 12; ⅓π · 25 · 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder and a cone have the same radius and height. The cylinder holds 600 mL. The cone holds

A. 300 mL

B. 200 mL

C. 600 mL

D. 1,800 mL

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q5/5

3136

QUESTION ID

S78-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 200 mL

GRADING NOTE ⅓.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • MAIN • Q5/5

SET 78 • HSG-GMD.A.3

BACKWARD 1

Area of a circle • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of a circle with radius 6?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q1/5

3138

QUESTION ID

S78-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Exact area of a circle with diameter 10?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q2/5

3139

QUESTION ID

S78-B1-Q2

TYPE

Short answer

CORRECT ANSWER

25π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q2/5

A base of radius 3 has area

A. 9

B. 9π

C. 3π

D. 6π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q3/5

3140

QUESTION ID

S78-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 9π

GRADING NOTE πr^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q3/5

True or false?

Squaring the radius comes before multiplying by π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q4/5

3141

QUESTION ID

S78-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Order of operations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q4/5

True or false?

πr^2 with $r = 3$ is $(3\pi)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q5/5

3142

QUESTION ID

S78-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 9π, not $9\pi^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 78 • BACKWARD 1 • Q5/5

SET 78 • HSG-GMD.A.3

BACKWARD 2

Find a missing side with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

A right triangle has hypotenuse 13 and one leg 5. Other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q1/5

3144

QUESTION ID

S78-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone has radius 6 and slant height 10. Height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q2/5

3145

QUESTION ID

S78-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q2/5

Radius, height, and slant height of a cone form a

A. square

B. right triangle

C. equilateral triangle

D. circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q3/5

3146

QUESTION ID

S78-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. right triangle

GRADING NOTE Height ⊥ base.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q3/5

True or false?

The slant height is longer than the height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q4/5

3147

QUESTION ID

S78-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q4/5

True or false?

The slant height can be used as the height in $V = \frac{1}{3}\pi r^2 h$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q5/5

3148

QUESTION ID

S78-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Use the perpendicular height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 78 • BACKWARD 2 • Q5/5

SET 78 • HSG-GMD.A.3

BACKWARD 3

Evaluate a formula and round • Grade 7 • Expressions & Equations • 7.EE.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate ⅓ · 36 · 8.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q1/5

3150

QUESTION ID

S78-B3-Q1

TYPE

Short answer

CORRECT ANSWER

96

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Round 90π to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q2/5

3151

QUESTION ID

S78-B3-Q2

TYPE

Short answer

CORRECT ANSWER

282.7

GRADING NOTE Rounding to 0.1 required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$96\pi \approx$

A. 301.6

B. 301.5

C. 3,016

D. 96.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q3/5

3152

QUESTION ID

S78-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 301.6

GRADING NOTE Use the π key.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q3/5

True or false?

Using 3.14 for π may change the last digit of a rounded answer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q4/5

3153

QUESTION ID

S78-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Use the π key.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\frac{1}{3} \cdot 100 \cdot 12 = 1{,}200$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q5/5

3154

QUESTION ID

S78-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 78 • BACKWARD 3 • Q5/5

SET 78 • HSG-GMD.A.3

BACKWARD 4

Which solid uses ⅓? • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

True or false?

Pyramids and cones use V = ⅓Bh.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q1/5

3156

QUESTION ID

S78-B4-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pointed solids.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which has the formula V = Bh: a cone or a cylinder?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q2/5

3157

QUESTION ID

S78-B4-Q2

TYPE

Short answer

CORRECT ANSWER

Cylinder

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$V = \frac{1}{3}\pi r^2 h$ is the volume of a

A. cylinder

B. cone

C. prism

D. sphere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q3/5

3158

QUESTION ID

S78-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. cone

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q3/5

Why does a cone have ⅓ of the matching cylinder's volume?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q4/5

3159

QUESTION ID

S78-B4-Q4

TYPE

Short answer

CORRECT ANSWER

Three cones fill the cylinder (the informal argument).

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q4/5

True or false?

A pyramid uses V = Bh.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q5/5

3160

QUESTION ID

S78-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE ⅓Bh.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B4-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 78 • BACKWARD 4 • Q5/5

SET 78 • HSG-GMD.A.3

FORWARD 1

Composite and hollow solids • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A silo is a cylinder (radius 4 m, height 10 m) topped by a cone (radius 4 m, height 3 m). Find its exact volume.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q1/5

3162

QUESTION ID
S78-F1-Q1

TYPE
Short answer

CORRECT ANSWER
176π m^3

GRADING NOTE Form required: exact value. $160\pi + 16\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A pipe is a cylinder of length 20 cm with outer radius 5 cm and inner radius 4 cm. Find the volume of the material, exact.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q2/5

3163

QUESTION ID
S78-F1-Q2

TYPE
Short answer

CORRECT ANSWER
180π cm^3

GRADING NOTE Form required: exact value. $\pi(25 - 16)(20)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q2/5

True or false?

The volume of a composite solid is the sum of the volumes of its non-overlapping parts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q3/5

3164

QUESTION ID

S78-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Additivity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A cylinder (V = 300) has a cone with the same base and height carved out. Remaining volume?

A. 300

B. 200

C. 0

D. 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q4/5

3165

QUESTION ID

S78-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 200

GRADING NOTE 300 − 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A pencil tip is a cone of radius 0.4 cm and height 1.5 cm on a cylinder of radius 0.4 cm and length 15 cm. Find the total volume to the nearest tenth.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q5/5

3166

QUESTION ID
S78-F1-Q5

TYPE
Short answer

CORRECT ANSWER
7.8 cm³

GRADING NOTE Rounding to 0.1 required. 2.4π + 0.08π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 78 • FORWARD 1 • Q5/5

SET 78 • HSG-GMD.A.3

FORWARD 2

Mass from volume and density • High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A steel cylinder has radius 2 cm and height 5 cm. Steel has density 7.85 g/cm³. Find the mass to the nearest gram.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q1/5

3168

QUESTION ID
S78-F2-Q1

TYPE
Short answer

CORRECT ANSWER
493 g

GRADING NOTE Rounding to the nearest gram required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q1

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cone of ice cream (radius 3 cm, height 10 cm) has density 0.55 g/cm³. Find the mass to the nearest gram.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q2/5

3169

QUESTION ID
S78-F2-Q2

TYPE
Short answer

CORRECT ANSWER
52 g

GRADING NOTE Rounding to the nearest gram required. 30π cm³.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q2

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q2/5

True or false?

Mass = volume ÷ density.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q3/5

3170

QUESTION ID

S78-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Mass = density × volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q3

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q3/5

A pyramid of volume 400 cm³ is made of material with density 2 g/cm³. Mass?

A. 200 g

B. 402 g

C. 1,600 g

D. 800 g

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q4/5

3171

QUESTION ID

S78-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 800 g

GRADING NOTE 2 × 400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q4

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A gold cube has volume 8 cm³ and mass 154.4 g. Find the density.

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q5/5

3172

QUESTION ID
S78-F2-Q5

TYPE
Short answer

CORRECT ANSWER
19.3 g/cm³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q5

High School — Geometry • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 78 • FORWARD 2 • Q5/5

SET 79 • HSG-GMD.A.3 • GEOMETRIC MEASUREMENT AND DIMENSION

Use the volume formula for spheres, including composite solids

MAIN	Geometry main question	**HSG-GMD.A.3**
BACKWARD 1	Cubes of numbers	**6.EE.A.1**
BACKWARD 2	Radius from diameter	**7.G.B.4**
BACKWARD 3	Multiply fractions by numbers	**7.NS.A.2**
BACKWARD 4	Cylinder and cone volumes	**HSG-GMD.A.3**
FORWARD 1	Scaling spheres	**HSG-GMD.A.3**
FORWARD 2	Model real objects as spheres	**HSG-MG.A.1**

SET 79 • HSG-GMD.A.3

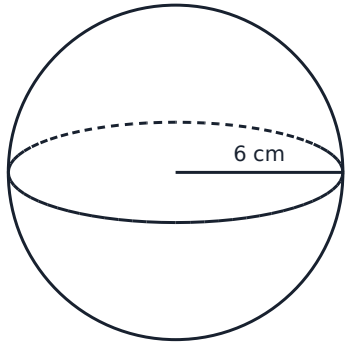
MAIN

Geometry main question • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Find the volume of a sphere with radius 6 cm, exact and to the nearest tenth.

Volume: _____



Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q1/5

3175

QUESTION ID

S79-M-Q1

TYPE

Short answer

CORRECT ANSWER

288π cm³ ≈ 904.8 cm³

GRADING NOTE Form required: both, with units. 4/3 π · 216.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the volume of a ball with diameter 10 in., exact and to the nearest tenth.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q2/5

3176

QUESTION ID

S79-M-Q2

TYPE

Short answer

CORRECT ANSWER

$500\pi/3$ in^3 ≈ 523.6 in^3

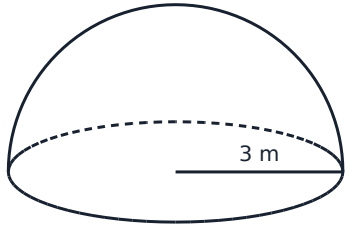
GRADING NOTE Form required: both, with units. r = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S79-M-Q2**

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the volume of a hemisphere with radius 3 m, exact and to the nearest tenth.



Volume: _____

Not drawn to scale.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q3/5

3177

QUESTION ID

S79-M-Q3

TYPE

Short answer

CORRECT ANSWER

18π m³ ≈ 56.5 m³

GRADING NOTE Form required: both, with units. ½ · 4/3 π · 27.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A capsule is a cylinder of radius 3 cm and length 10 cm with a hemisphere on each end. Find its volume, exact and to the nearest tenth.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q4/5

3178

QUESTION ID
S79-M-Q4

TYPE
Short answer

CORRECT ANSWER
126π cm³ ≈ 395.8 cm³

GRADING NOTE Form required: both, with units. 90π + 36π (two hemispheres make one sphere).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q4/5

An ice cream cone (radius 2, height 9) is topped by a hemisphere of radius 2. The total exact volume is

A. 52π/3

B. 28π

C. 12π

D. 44π/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q5/5

3179

QUESTION ID

S79-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 52π/3

GRADING NOTE 12π + 16π/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • MAIN • Q5/5

SET 79 • HSG-GMD.A.3

BACKWARD 1

Cubes of numbers • Grade 6 • Expressions & Equations • 6.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

$6^3 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q1/5

3181

QUESTION ID

S79-B1-Q1

TYPE

Short answer

CORRECT ANSWER

216

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$5^3 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q2/5

3182

QUESTION ID

S79-B1-Q2

TYPE

Short answer

CORRECT ANSWER

125

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$3^3 =$

A. 9

B. 81

C. 6

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q3/5

3183

QUESTION ID

S79-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 27

GRADING NOTE $3 \cdot 3 \cdot 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$2^3 = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q4/5

3184

QUESTION ID

S79-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Cube.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$4^3 = 12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q5/5

3185

QUESTION ID

S79-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 64.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 79 • BACKWARD 1 • Q5/5

SET 79 • HSG-GMD.A.3

BACKWARD 2

Radius from diameter • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A ball has diameter 10. Radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q1/5

3187

QUESTION ID

S79-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q1/5

True or false?

Using the diameter in $4/3\ \pi r^3$ makes the answer 8 times too large.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q2/5

3188

QUESTION ID

S79-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 2^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q2/5

A sphere with diameter 12 has r^3 =

A. 216

B. 144

C. 36

D. 1,728

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q3/5

3189

QUESTION ID

S79-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 216

GRADING NOTE 6^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A sphere has radius 2.5. Diameter?

Diameter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q4/5

3190

QUESTION ID

S79-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q4/5

True or false?

The radius of a sphere is twice its diameter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q5/5

3191

QUESTION ID

S79-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 79 • BACKWARD 2 • Q5/5

SET 79 • HSG-GMD.A.3

BACKWARD 3

Multiply fractions by numbers • Grade 7 • The Number System • 7.NS.A.2

ANSWER KEY Reduced copy of the question page, with its answer

$4/3 \cdot 216 =$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q1/5

3193

QUESTION ID

S79-B3-Q1

TYPE

Short answer

CORRECT ANSWER

288

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

½ · 4/3 · 27 =

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q2/5

3194

QUESTION ID

S79-B3-Q2

TYPE

Short answer

CORRECT ANSWER

18

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$4/3 \cdot 125 =$

A. 375/4

B. 500/3

C. 125/3

D. 500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q3/5

3195

QUESTION ID

S79-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 500/3

GRADING NOTE Numerators.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$4/3 \cdot 27 = 36$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q4/5

3196

QUESTION ID

S79-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $27 \div 3 \times 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$4/3 \cdot 6 = 24$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q5/5

3197

GRADING NOTE 8.

QUESTION ID

S79-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 79 • BACKWARD 3 • Q5/5

SET 79 • HSG-GMD.A.3

BACKWARD 4

Cylinder and cone volumes • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Cylinder radius 3, length 10: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q1/5

3199

QUESTION ID

S79-B4-Q1

TYPE

Short answer

CORRECT ANSWER

90π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Cone radius 2, height 9: exact volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q2/5

3200

QUESTION ID

S79-B4-Q2

TYPE

Short answer

CORRECT ANSWER

12π

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B4-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 79 • BACKWARD 4 • Q2/5

3200